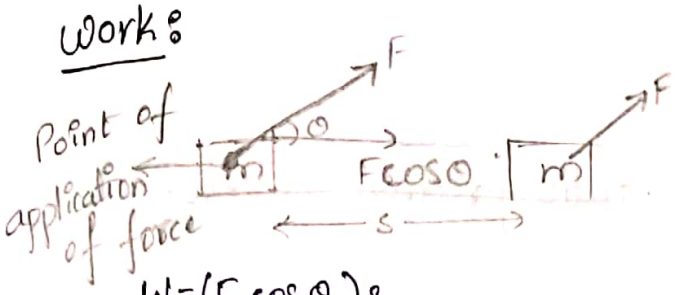
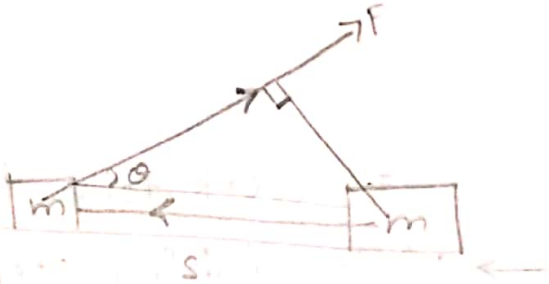
Work, Power & Energy

Work
- Not considered in Physics: Mental work
- physical work ✓

Work:





Point of application of force

$W = (F\cos\theta)s$

$W = Fs\cos\theta$

$\Rightarrow \boxed{W = \vec{F}\cdot\vec{s}}$

s = displacement of point of application of force.

* It is a scalar quantity.
* S.I unit — Joule
* C.G.S unit — erg

} relation $\Rightarrow 1\ J = 10^7\ erg$

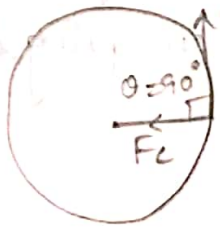
* work can be +ve, −ve, 0.

$W = +ve \Rightarrow \cos\theta = +ve \Rightarrow 0 \le \theta < 90°$

$W = -ve \Rightarrow \cos\theta = -ve \Rightarrow 90 < \theta \le 180°$

$W = 0 \Rightarrow \cos\theta = 0 \Rightarrow F = 0$ or $s = 0$ or $\theta = 90°$.

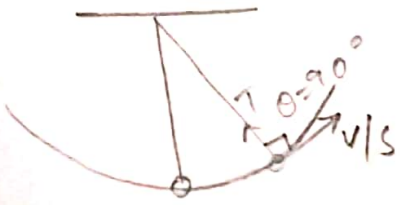
→



uniform circular motion.

Work done by F_c is always zero.
(also in case of Non-U.C.M)

→

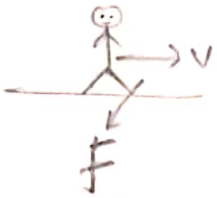


during oscillation,

$\boxed{W_T = 0}$ work done by Tension force is zero.

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→



while man walking / running on ground without slipping



$f = F$ on ground by man

point of application of force.

⇒ work done by static friction is zero.

→ while climbing steps

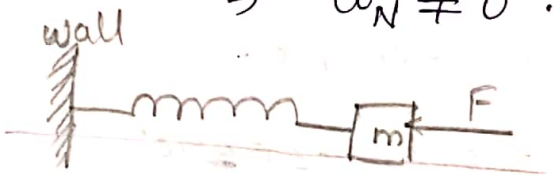


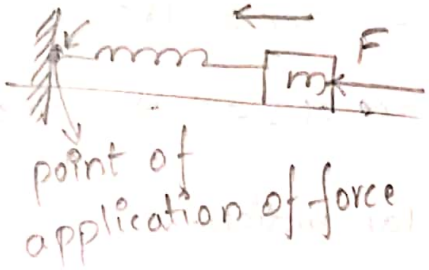
⇒ work done by normal reaction force is zero.

* when there is oil on the ground then there will be friction force & (displacement) will not be at rest.

⇒ $W_N \neq 0$.

→





during compression, work done by the force applied by wall on spring is zero.

Work calculation formulas:

① $W = \bar{F} \cdot \bar{S}$

$\bar{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$

$\bar{S} = S_x\hat{i} + S_y\hat{j} + F_z\hat{k}$

$$W = F_x S_x + F_y S_y + F_z S_z$$

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② ** $W = \int_{r_i}^{r_f} \bar{F} \cdot d\bar{r}$ $\qquad d\bar{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$

* $\bar{F}$  (for example)
- const ($\bar{F} = 2\hat{i} + 3\hat{j} + 4\hat{k}$)
- as a function of (x, y, z)
- as a function of (t)
- as a function of (v).

→ ① $\bar{F} = x^2y\,\hat{i} + yz\,\hat{j} + xyz\,\hat{k}$

$$W = \int \bar{F} \cdot d\bar{r}$$

$$= \int (x^2y\,\hat{i} + yz\,\hat{j} + xyz\,\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$= \int x^2y \cdot dx + yz \cdot dy + xyz\,dz$$

$$W = \int x^2y\,dx + \int yz \cdot dy + \int xyz \cdot dz$$

we can calculate work done with additional information in given question.

② $F = \frac{K}{v}\hat{i}$

$$W = \int \bar{F} \cdot d\bar{x} = \int \frac{K}{v} \cdot dx \; ; \quad v = \frac{dx}{dt}$$

$$W = \int \frac{K}{dx} \cdot dt \cdot dx = \int K \cdot dt$$

$$\boxed{W = Kt + C}$$

③ $F = 2t\,\hat{i}$

$$W = \int 2t \cdot dx = 2\int t \cdot dx$$

$$F = 2t \Rightarrow ma = 2t \Rightarrow a = \frac{2t}{m} \Rightarrow \frac{dv}{dt} = \frac{2t}{m}$$

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$$\int_0^v dv = \int_0^t \frac{2t}{m}\,dt$$

$$v = \frac{2}{m}\cdot\frac{t^2}{2}$$

$$\boxed{v = t^2/m}$$

$$\frac{dx}{dt} = \frac{t^2}{m} \Rightarrow dx = \frac{t^2}{m}\,dt$$

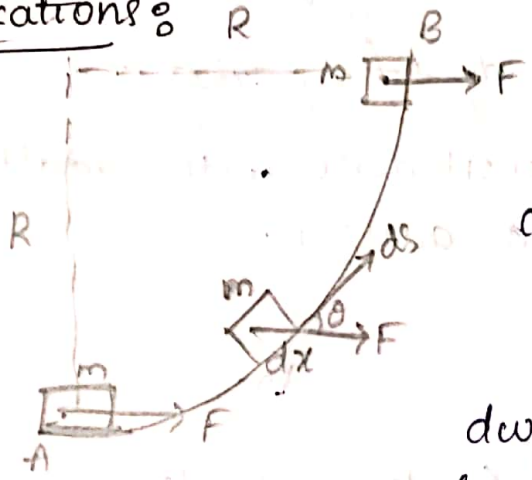
$$W = \int_0^t 2t\cdot\frac{t^2}{m}\,dt$$

$$= \frac{2}{m}\left(\frac{t^4}{4_2}\right) + C$$

$$\boxed{W = \frac{t^4}{2m} + C}$$

Applications:

①



$$\cos\theta = \frac{dx}{ds}$$

$$\Rightarrow dx = ds\cos\theta$$

$$dw = F\,ds\cos\theta$$

$$\int dw = \int F\,dx$$

$$w = \int_0^R F\,dx$$

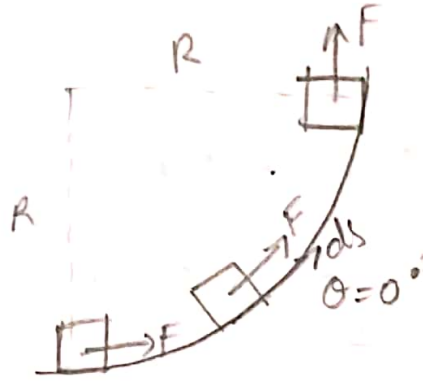
$$= F[x]_0^R$$

$$\boxed{w = FR}$$

work done by mass (m).

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(2)



'constant force F' tangential to curve.

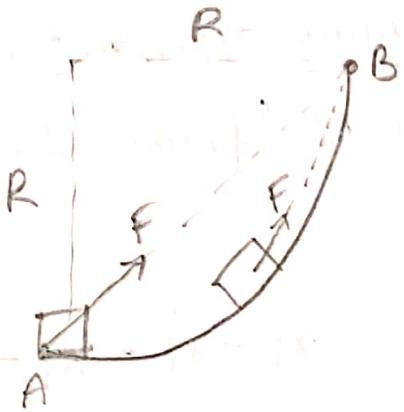
$dW = F\,ds\cos\theta$

$\int dW = \int F\,ds$

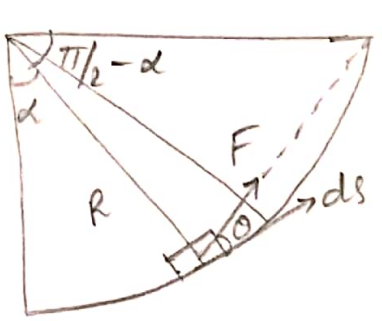
$W = \int F\cdot ds = F\int ds = F$ (total arc length)

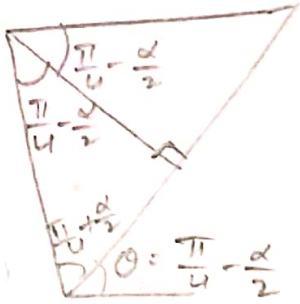
$$\boxed{W = F\pi R/2}$$

(3)



constant force 'F' aiming opposite point 'B'.





$dW = F\,ds\cos\theta = F(R\,d\alpha)\cos\left(\frac{\pi}{4}-\frac{\alpha}{2}\right)$

$= \frac{FR}{\sqrt{2}}\left(\cos\frac{\alpha}{2}+\sin\frac{\alpha}{2}\right)\cdot d\alpha$

$$W = \int_{\alpha=0}^{\pi/2} dW = \frac{FR}{2}\left[\int_0^{\pi/2}\cos\frac{\alpha}{2}\cdot d\alpha + \int_0^{\pi/2}\sin\frac{\alpha}{2}\cdot d\alpha\right]$$

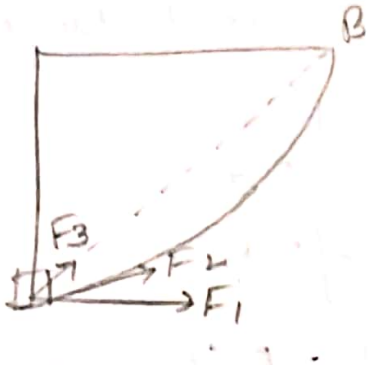
$$W = \frac{FR}{\sqrt{2}}\left[\frac{\sin\frac{\alpha}{2}}{1/2} - \frac{\cos\alpha/2}{1/2}\right]_0^{\pi/2}$$

$$W = \sqrt{2}FR\left[\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - 0 + 1\right]$$

$$\boxed{W = \sqrt{2}FR}$$

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Conclusion:



$$W_{F_1} = F_1 R = F_1 R$$

$$W_{F_2} = \frac{F_2 \pi R}{2} = F_2\left(\frac{\pi}{2}R\right)$$

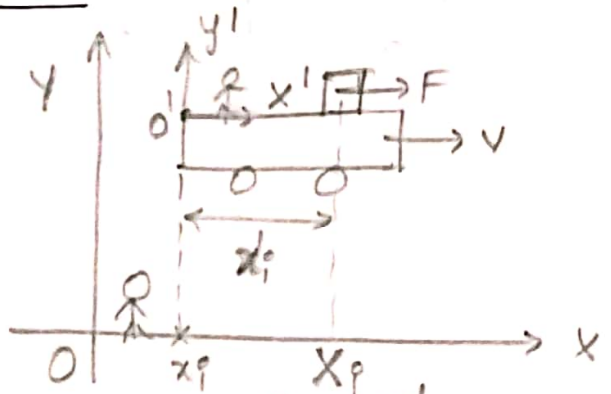
$$W_{F_3} = \sqrt{2} F_3 R = F_3\left(\sqrt{2}R\right)$$

Total work done $= W_{F_1} + W_{F_2} + W_{F_3}$

*** Work done is a 'frame dependent quantity'!

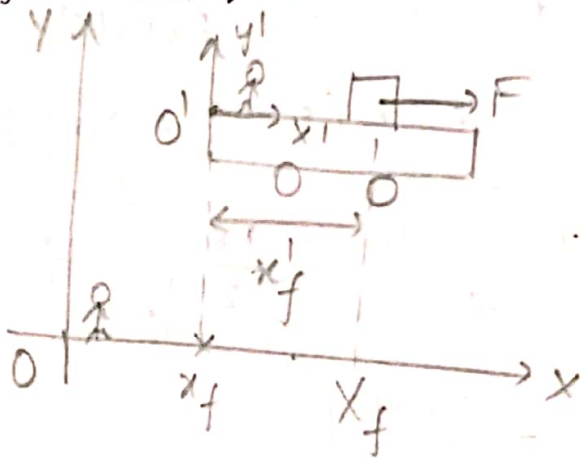
Bcoz, displacement depends of 'frame of reference'

Explanation:-



$$x_i' + x_i'^{\circ} = X_i^{\circ} \longrightarrow (1)$$

After time 't'. ground



$$x_f + x_f' = X_f \longrightarrow (2)$$

w.r.to observer O',

$$W' = F(x_f' - x_i'^{\circ})$$

w.r.to observer O,

$$W = F(X_f - X_i^{\circ}) = F(x_f + x_f' - x_i^{\circ} - x_i'^{\circ})$$

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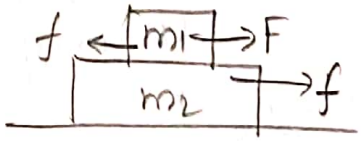
$$W = F(x_f - x_i) + F(x'_f - x'_i)$$

$$W = F\,S_{\text{trolley}} + \#\,\omega' \text{ w.r.to } O.$$

***work done by static friction may be "positive or negative" but not work done by static friction is always zero'.

Explanation :-

* for two blocks → 'S'



$\mu = 0$

* common acceleration:

$$a = \frac{F}{m_1+m_2}$$

$$f = m_2 a$$

$$f = \frac{m_2 F}{m_1+m_2}$$

* For $F \le F_{max}$ block move together.

$f \le f_s$

$$\frac{m_2 F}{m_1+m_2} \le \mu m_1 g$$

$$F \le \mu\left(\frac{m_1}{m_2}\right) g(m_1+m_2)$$

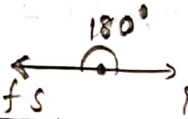
$$\Rightarrow F_{max} = \mu\left(\frac{m_1}{m_2}\right) g(m_1+m_2)$$

* S = displacement in time 't' -

$$S = \frac{1}{2} a t^2$$

* work done by 'fs' on m_1 -

$$W_1 = -f_s(s)$$



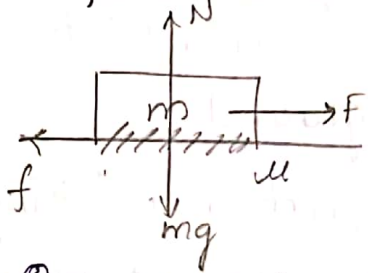
$$\boxed{W_1 = -f_s(s)}$$

* work done by f_s on m_2 -

$$W_2 = (f_s)(s)$$

$$\boxed{W_2 = f_s(s)}$$

*for one block



① $F < f_s$; block is at rest

$\therefore f = f_{static} = F$

Work done by static friction $= f(s) = f(0) = 0$

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* work done by f_s on system -

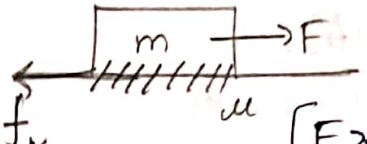
$$W_0 = W_1 + W_2$$

$$\boxed{W = 0}$$

*** workdone by "kinetic friction" may be +ve & -ve but net work done by kinetic friction is always "-ve".

Explanation:-

for one block → 's'



$[F > f_d \quad F > \mu mg]$

* If $F > f_d$, block slides on ground

$\therefore$ friction is "kinetic"; $\boxed{f_K = \mu_K mg}$

* acceleration of block, $a = \frac{F - f_K}{m}$.

* displacement of block in t sec, $s = \frac{1}{2}at^2$

* work done by "kinetic friction" is,

$$W_1 = -f_K(s) = -(\mu_K mg)s$$

* work done by applied force,

$$W_2 = F(s) = (ma + f_K)s$$

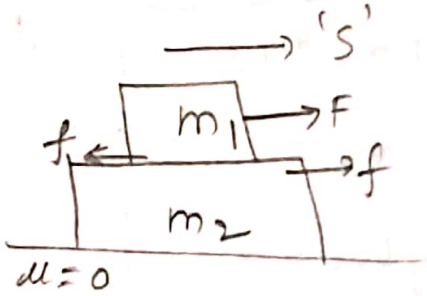
* net work done by all forces,

$$W = W_1 + W_2 = -f_K(s) + ma(s) + f_K(s)$$

$$\boxed{W = ma(s)}$$

for two blocks

* If $F > F_{max}$ then blocks move with seperate accelerations.



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$a_1 = \frac{F - f_k}{m_1} = \frac{F - \mu m_2 g}{m_1}$, $a_2 = \frac{f_k}{m_2} = \frac{\mu_k m_1 g}{m_2}$

* work done by f_k on m_1 is,

$$\boxed{W_1 = -f_k(s_1)} \qquad s = \frac{1}{2} a_1 t^2$$

* work done by f_k on m_2 is,

$$\boxed{W_2 = f_k(s_2)} \qquad s_2 = \frac{1}{2} a_2 t^2$$

* work done by f_k on system,

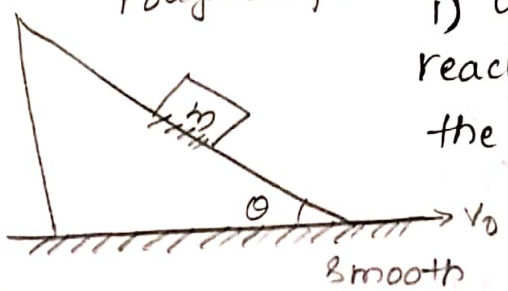
$$W = W_1 + W_2$$

$$= -f_k(s_1) + f_k(s_2)$$

$$= f_k(s_2 - s_1) \qquad (s_1 > s_2)$$

$$\boxed{W = -f_k(s_{rel})}$$

Applications :

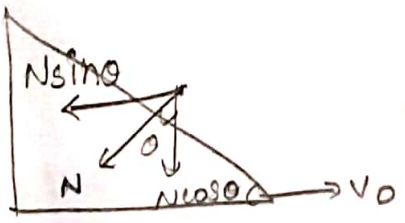


1) work done by normal reaction in 1st 't' secs on wedge, If the block is at rest on wedge.

block is at rut.

$\Rightarrow f = mg\sin\theta$

$N = mg\cos\theta$.



$s = v_0 t$

$W = \vec{F} \cdot \vec{s}$

$W = -(N\sin\theta)(v_0 t)$

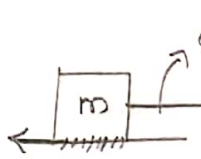
$= -mg\cos\theta\sin\theta v_0 t$

$$\boxed{W = -\frac{mgv_0 t}{2}\sin 2\theta}$$

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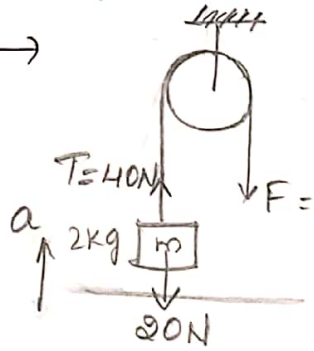
2) work done by Normal rxn on block,

$$W' = \frac{mgvot}{2}\sin\theta$$

→ 

work done → +ve.

f

→ 

work done by ① T.
② gravity
③ net force
④ in first 2 sec.

$$a = \frac{40-20}{2} = \frac{20}{2} = 10\,m/s^2$$

② work done by Gravity is,

$$W_g = (mg)s\cos 180$$

$$= 20 \times 8(-1)$$

$$\boxed{W_g = -400J}$$

$$s = \frac{1}{2}(10)(2)^2$$

$$= 20$$

① work done by T is,

$$W_T = Ts\cos(0)$$

$$= 40 \times 20 \times 1$$

$$\boxed{W_T = 800\,J}$$

③ Work done by net force,

$$W_{nf} = -400 + 800$$

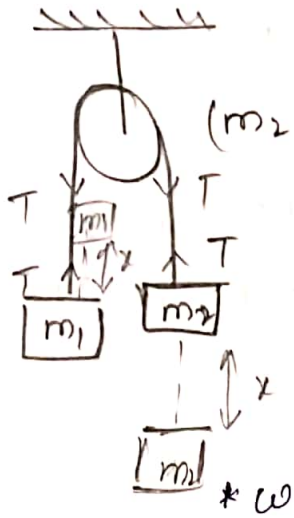
$$= 400\,J$$

or

$$W_{nf} = mas = 2 \times 10 \times 20 = 400J$$

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→



$$a = \frac{(m_2 - m_1)g}{m_1 + m_2}$$

In first 't' secs

$$x = \frac{1}{2}at^2$$

* work done by gravity on m_1 –

$$W = -m_1 g(x) = -m_1 (g)\left(\frac{1}{2}at^2\right)$$

$$\boxed{W_1 = -\frac{m_1 g^2}{2}\left(\frac{m_2 - m_1}{m_1 + m_2}\right)t^2}$$

* work done against the gravity,

$$W = -(-m_1 g x) = m_1 g x$$

* work done by gravity on m_2 –

$$\boxed{W_2 = \frac{m_2 g^2}{2}\left(\frac{m_2 - m_1}{m_1 + m_2}\right)t^2}$$

* Net work done on system.

$$W = W_1 + W_2 = \frac{(m_2 - m_1)^2}{m_1 + m_2} \times \frac{g^2 t^2}{2}$$

$$\boxed{W = \frac{1}{2} g^2 t^2 \left(\frac{(m_2 - m_1)^2}{m_1 + m_2}\right)}$$

* work done by 'T' on m_1,

$$\boxed{W_{T_1} = Tx}$$

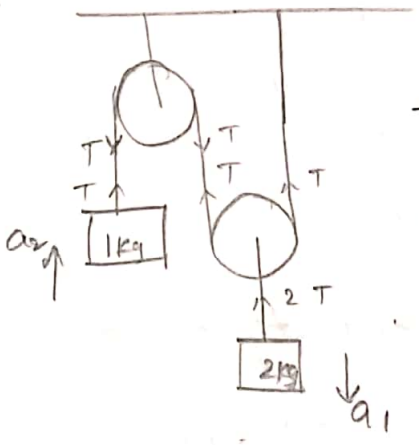
* work done by 'T' on m_2,

$$\boxed{W_{T_2} = -Tx}$$

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∴ net work done by Tension,

$$W = Tx - Tx = 0.$$

→ 

Find work done by gravity & tension on both blocks.

① $\Sigma \bar{T} \cdot \bar{a} = 0$

$\Rightarrow -2Ta_1 + Ta_2 = 0$

$\boxed{a_2 = 2a_1} \rightarrow ①$

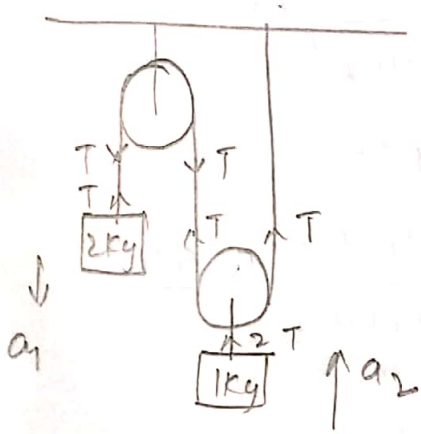
$2g - 2T = 2a_1 \Rightarrow g - T = a_1 \rightarrow ②$

$T - g = a_2 \quad - ③$

$a_1 + a_2 = 0$

$3a_1 = 0 \Rightarrow \boxed{a_1 = 0} \Rightarrow \boxed{a_2 = 0}$

* $T = g \cdot \quad \omega_g = 0, \ \omega_T = 0$

→ 

① $\Sigma \bar{T} \cdot \bar{a} = 0$

$-Ta_1 + 2Ta_2 = 0$

$\boxed{a_1 = 2a_2} \rightarrow ①$

$2g - T = 2a_1$

$2(g - a_1) = T \rightarrow ②$

$2T - g = a_2$

$4g - 4a_1 - g = a_2$

$3g - 4a_1 = a_2$

$3g - 8a_2 = a_2$

$\boxed{a_2 = g/3}$

$\Rightarrow \boxed{a_1 = 2g/3}$

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(2) c) $2g - \frac{4g}{3} = T$

$$\boxed{T = \frac{2g}{3}}$$

* workdone by gravity on 1kg is,

$$W_{g_1} = -(1g)(s_2)$$

$$= -g\left(\frac{1}{2}a_2t^2\right)$$

$$= -g\left(\frac{1}{2}\right)\left(\frac{g}{3}\right)t^2$$

$$W_{g_1} = -\frac{g^2t^2}{6}$$

* work done by gravity on 2kg is,

$$W_{g_2} = (2g)s_1$$

$$= 2g\left(\frac{1}{2}a_1t^2\right)$$

$$W_{g_2} = \frac{2g^2t^2}{3}$$

*work done by Tension on 1kg is,

$$W_{T_1} = (2T)s_2$$

$$= 2\left(\frac{2g}{3}\right)\left(\frac{1}{2}\right)\left(\frac{g}{3}\right)t^2$$

$$= \frac{2g^2t^2}{9}$$

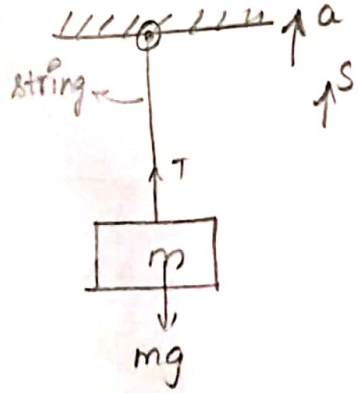
* work done by tension 2kg is,

$$W_{T_2} = -Ts_1$$

$$= -\left(\frac{2g}{3}\right)\left(\frac{1}{2}\right)\left(\frac{2g}{3}\right)t^2$$

$$= -\frac{2g^2t^2}{9}$$

→



$T - mg = ma$

$T = m(g+a)$

what is work done by gravity on block,

$$\boxed{W_g = -mgs}$$

what is work done by Tension on block,

$$W_T = Ts = m(g+a)s$$

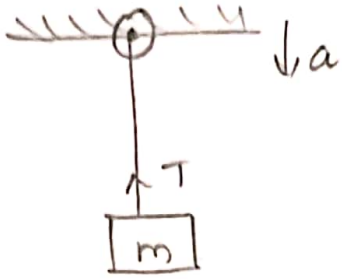
work done by Net force,

$$W = W_T + W_g = -mgs + m(g+a)s$$

$$= (ma)s.$$

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→



$T = m(g-a)$

$w_g = mgs$

$w_T = -Ts$

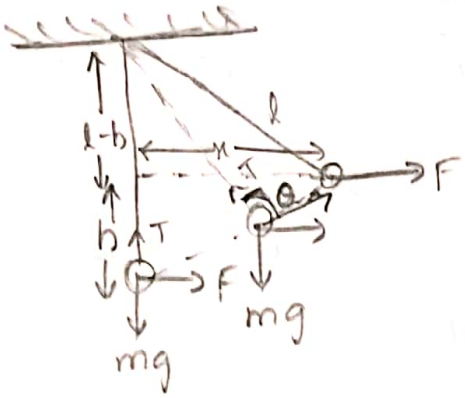
$= -m(g-a)s$

$w = w_g + w_T$

$= mgs - mgs + mas$

$\boxed{w = (ma)s}$

→



$\cos\theta = \frac{l-h}{l}$

$\boxed{h = l(1-\cos\theta)}$

$\sin\theta = x/l$

$\Rightarrow x = l\sin\theta$

* work done by Tension

$w_T = T\cos 90^\circ = 0$

* work done by gravity

$w_g = -mgh$

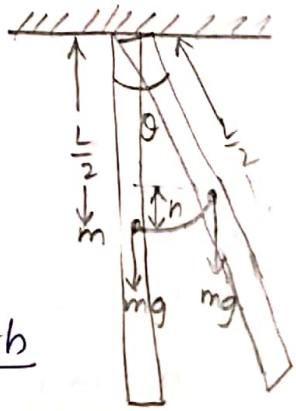
$= -mgl(1-\cos\theta)$

$w_g = mgl(\cos\theta - 1)$

* work done by F.

$w_F = Fx = Fl\sin\theta$.

→





work done by gravity

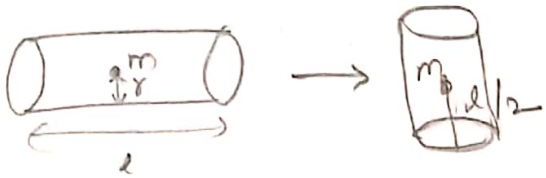
$w_g = -mgh$

$= -mg\frac{l}{2}(1-\cos\theta)$

$\cos\theta = \frac{l/2 - h}{l/2}$

$\boxed{h = l/2(1-\cos\theta)}$

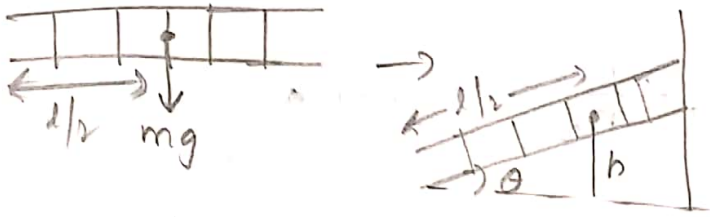
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work done by gravity when tilting as shown.

$$W_g = -mgh$$

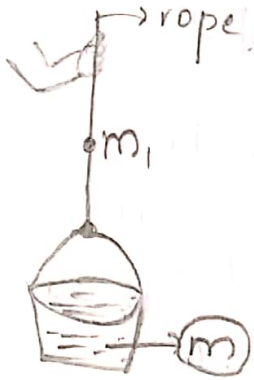
$$W_g = -mg\left(\frac{l}{2} - r\right)$$



work done by gravity while as shown.

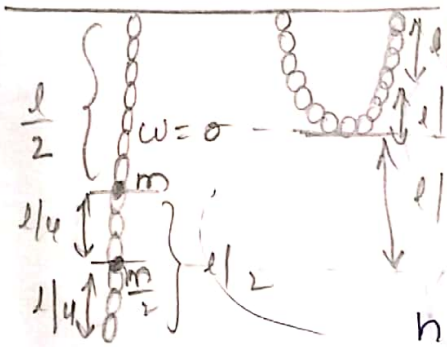
$$W_g = -mgh$$

$$= -mg\frac{l}{2}\sin\theta .$$



work done in pulling the rope-bucket system slowly,

$$W = mL + m_1 g\frac{L}{2}$$



work done by gravity in lifting the lower end of chain to point of suspension.

$$h = \frac{l}{4} + \frac{l}{4} = \frac{l}{2}$$

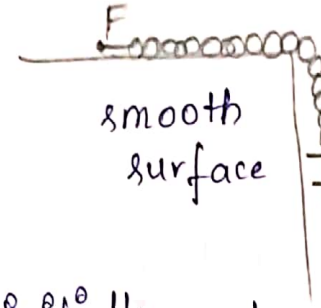
$$W_g = -\frac{m}{2}gh$$

$$= -\frac{m}{2}g\left(\frac{l}{2}\right)$$

$$\boxed{W_g = -mgl/4}$$

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[chain is pulled slowly]

→ 

work done in pulling the chain on to the table.

initially, $\frac{1}{n_1}$th part of chain is hanging & finally, $\frac{1}{n_2}$th part of chain is hanging.

$\Rightarrow l_1 = \frac{L}{n_1}, \quad l_2 = \frac{L}{n_2}$

Here $n_1 > 1$, $n_2 > 1$, $n_1 < n_2$

since $l_1 > l_2$

$$\frac{L}{n_1} > \frac{L}{n_2}$$

* small work done, $dw = (dm)gl$

$$dw = \left(\frac{m}{L} \cdot dl\right) g\, l$$

$$dw = \frac{mg}{L} \cdot l \cdot dl$$

$$w = \int_{l_2}^{l_1} dw = \frac{mg}{L} \int_{l_2}^{l_1} l \cdot dl$$

$$w = \frac{mg}{2L}(l_1^2 - l_2^2)$$

$$w = \frac{mg}{2L}\left(\frac{L^2}{n_1^2} - \frac{L^2}{n_2^2}\right)$$

$$w = \frac{mgL}{2}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

work done by gravity,

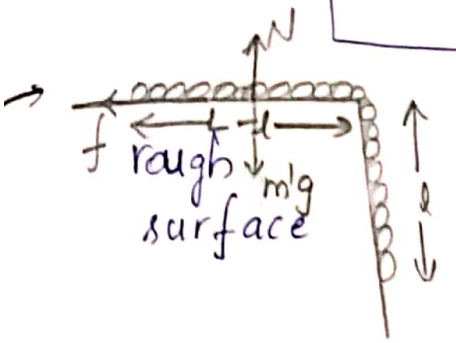
$$w = -\frac{mgL}{2}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

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* If chain is completely pulled on to the table,

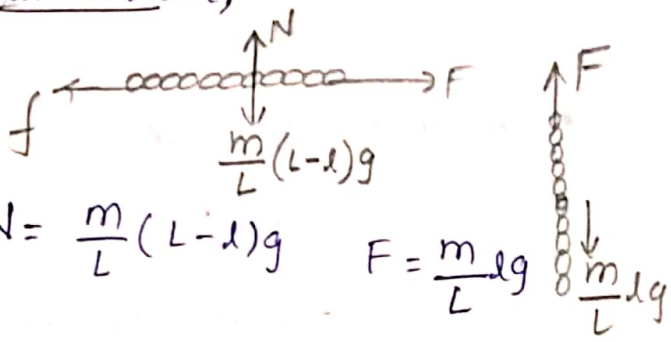
$l_2 = 0 \Rightarrow W = \frac{mgl_1^2}{2L} = \frac{mgL^2}{2Ln_1^2} = \frac{mgL}{2n_1^2}$

$$\boxed{W = \frac{mgl}{2n_1^2}}$$



$f_L = F \qquad l = \frac{L}{n}$

* when chain is in limiting equilibrium,



$\mu N = \frac{m}{L} lg$

$N = \frac{m}{L}(L-l)g \qquad F = \frac{m}{L} lg$

$\mu \frac{m}{L}(L-l)g = \frac{m}{L} lg$

$\mu L - \mu l = l$

$\mu L = (\mu+1) l$

$l = \left(\frac{\mu+1}{\mu}\right) l$

$$\boxed{l = \frac{\mu L}{\mu+1}}$$ → maximum length of chain hanging without slipping on the table.

* fractional length of chain hanging,

$$\boxed{\frac{l}{L} = \frac{\mu}{\mu+1}}$$

$\Rightarrow \frac{1}{n} = \frac{\mu}{\mu+1}$

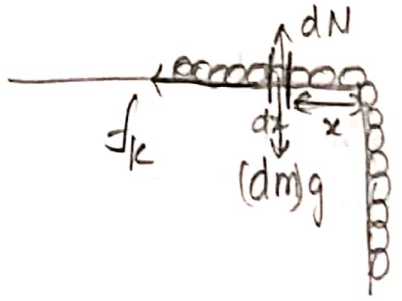
$\frac{\mu+1}{\mu} = n \Rightarrow 1 + \frac{1}{\mu} = n$

$1 - n = -\frac{1}{\mu}$

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$$\mu = \frac{1}{n-1} \quad (n>1)$$

* when chain starts sliding, calculation of workdone by kinetic friction.



* Small work done,

$$dw = -(f_k)\,x$$

$$= -\mu_k (dN) \cdot x$$

$$dw = -\mu_k x\,(dm)\,g$$

$$dw = -\mu_k x g \frac{m}{L} \cdot dx \cdot x$$

$$dw = -\frac{\mu_k m g}{L}\, x \cdot dx$$

* Total work done,

$$w = \int_0^{L-d} dw = \int_0^{L-d} -\frac{\mu_k mg}{L}\, x \cdot dx$$

$$w = -\frac{\mu_k mg}{L} \int_{x=0}^{L-d} x \cdot dx$$

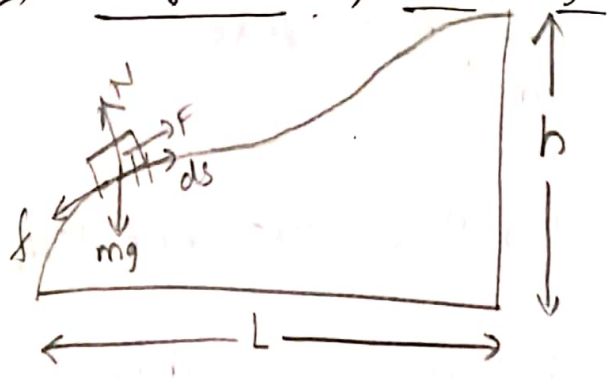
$$\boxed{w = -\frac{\mu_k mg}{2L}(L-d)^2}$$

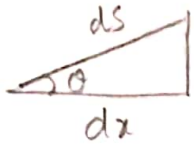
$L = \frac{l}{n}$

$\mu_k = \mu = \frac{1}{n-1}$

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→ Rough hill, block is dragged upto the hill.





$N = mg\cos\theta$

$dx = ds\cos\theta$

* work done by friction,

$$dw = -f_k(ds)$$

* Total work done by friction, $= -\mu_k N \cdot ds$

$$= -\mu_k mg(\cos\theta) \cdot ds$$

$$\boxed{dw = -\mu_k mg \cdot dx}$$

$$w = -\mu_k mg \int_{\text{bottom}}^{\text{top}} dx$$

$$\boxed{w = -\mu_k mg\, L} \quad ***$$

* work done by normal rxn,

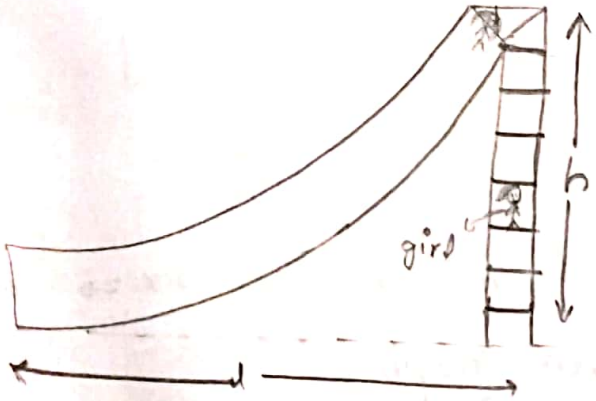
$$w_N = 0 \quad [\theta = 90^\circ]$$

* work done by gravity,

$$w_g = -mgh.$$

* work done by applied force,

→



while climbing,

* work done by ladder on girl $= 0$.

* work done by friction during sliding,

$$w = -f_k(s)$$

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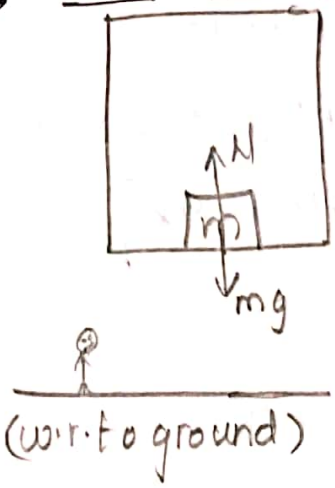
$W = -f_k (s)$ $\quad$ s – length of slide

(or) $\boxed{W = -\mu_k mgl}$

* work done by normal rxn is zero (while sliding).

* work done by gravity is $W_g = mgh$ (while sliding)

→ Block in accelerating lift:



w.r.to observer on ground,

① $W_N = Ns$

$= m(g+a)s$

$= m(g+a)\left(\frac{1}{2}at^2\right)$

② $W_g = -mgs = -\frac{mgat^2}{2}$

③ $W = W_N + W_g$

$= m(g+a)s - mgs$

$= mas = \frac{ma^2t^2}{2}$

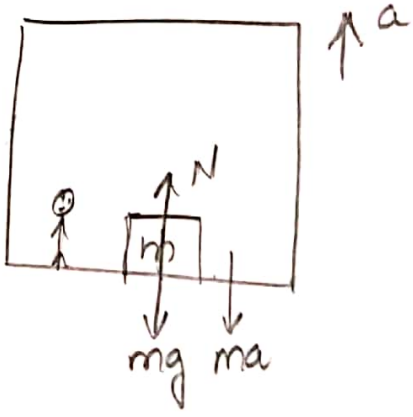
w.r.to observer in lift frame,

① $W_N = N(0) = 0$

$W_g = mg(0) = 0$

$W_{Pseudo} = ma(0) = 0$

* $\boxed{W = 0}$



→ For block moves from $x = -2$ to $x = 0$ as shown in the given graph.

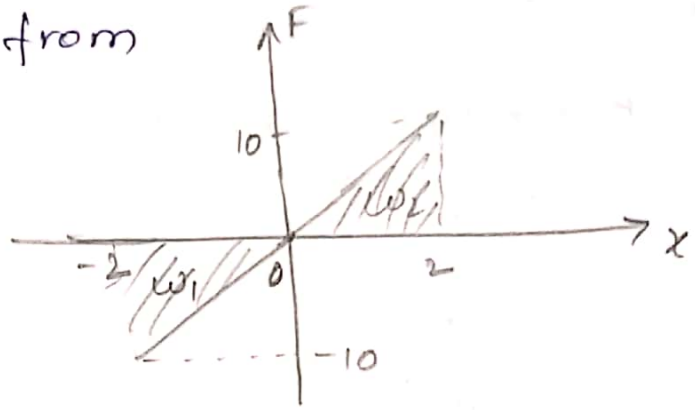
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for block moves from

$x = -2$ to 0

displacement $= 2m$

& force $= -ve$.



$$\omega_1 = -\frac{1}{2} \times 10 \times 2 = -10\,J$$

for block moves from

$x = 0$ to 2 $\Rightarrow$ displacement $= 2m$ &

force $= +ve$. $\omega_2 = \frac{1}{2} \times 10 \times 2 = 10\,J$

$\omega_T = \omega_1 + \omega_2 = -10 + 10 = 0$.

$W = mgh\left(1 - \frac{dl}{ds}\right)$ ↳ S.D

Energy:

The capacity to do work is called energy.

Work is done at the expense of energy.

- Energy
  - Mechanical energy
    - Kinetic energy
    - potential energy
  - heat energy
  - light energy
  - electric & magnetic energy
  - Sound energy

Energy → Joule
erg
calories
electronvolt
kilowatt hour

Kinetic Energy:

Energy possessed by a body during motion is called kinetic motion.

$1\,J = 10^7$ erg

$1\,cal = 4.18\,J$

$1\,eV = 1.6 \times 10^{-19}\,J$

$1\,Kwh = 10^3\,wh\,r$

$1\,Kwh = 3.6 \times 10^5\,J$

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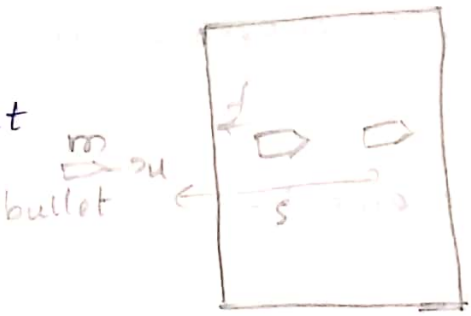
Explanation: explanation!

work done by the bullet

$= -F \cdot s$

$= -mas$

$= -m\left(\frac{0^2 - v^2}{2s}\right) \cdot s$



$W = \frac{1}{2}mv^2$

$\boxed{K.E = \frac{1}{2}mv^2}$

In general, $K.E = \frac{1}{2}mv^2 = \frac{1}{2}m(\vec{v}\cdot\vec{v})$.

relation b/w momentum & K.E :-

$K = \frac{1}{2}mv^2 = \frac{(mv)^2}{2m} = \frac{p^2}{2m}$.

$K \propto p^2$ when m is constant.

% of change in K.E $= \frac{K_f - K_i}{K_i} \times 100$

$= \frac{p_f^2 - p_i^2}{p_i^2} \times 100$

$\frac{\Delta K}{K} = \frac{2p \cdot \Delta p}{p^2}$

$\boxed{\frac{\Delta K}{K} \times 100 = 2 \cdot \frac{\Delta p}{p} \times 100}$ → use this when change is less than 5%.

→ $\frac{\Delta p}{p} \times 100 = 3$. Find $\frac{\Delta K}{K}$.

$\frac{\Delta K}{K} \times 100 = 2 \times 3 = 6\%$.

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→ If momentum increases 11%. find % change in K.E.

let $P_i = 100$

$P_f = 111$.

$$\frac{\Delta K}{K} \times 100 = \frac{(111-100)(111+100)}{(100)^2} \times 100$$

$$= \frac{211 \times 11}{100} = 23.21\ \%$$

→ If momentum decreases by 10% · find % change in K.E.

$$\frac{\Delta K}{K} \times 100 = \frac{(90)^2 - (100)^2}{(100)^2} \times 100$$

$$= \frac{190 \times (-19)}{100}$$

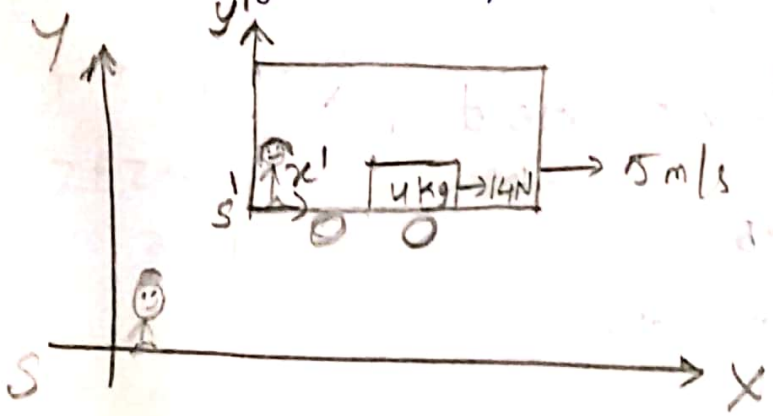
$$= -19\ \%$$

Kinetic energy is always +ve & zero.

* K.E is a scalar quantity.

* K.E is a frame dependent quantity.

→



Initially block is at rest w.r.t trolley (smooth)

* Force $F = 14N$ acts for $t = 3$ sec on block.

w.r.to observer S :-

$$\bar{V}_b = \bar{V}_{b|T} + \bar{V}_T$$

* initial velocity $u_i = 0 + 5 = 5\ m/s$.

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* initial K.E

$$K_i = \frac{1}{2} m u_i^2 = \frac{1}{2} \times 4 \times 5 \times 5 = 50\,J$$

* w.r.to observer S' :-

$$\bar{u}_{b|T} = 0$$

$$K_i = \frac{1}{2} m (u_{bT})^2$$

$$= 0.$$

* acceleration of block w.r.to trolley,

$$a = \frac{F}{m} = \frac{14}{4} = 3.5\,m/s^2.$$

* velocity of block w.r.to trolley at $t = 3$ sec

$$v_{b|T} = u_{b|T} + a_{b|T} t$$

$$= 0 + 3.5 \times 3$$

$$= 10.3$$

* K.E of block w.r.to S',

$$K_{b|T} = \frac{1}{2} m (v_{b|T})^2$$

$$= \frac{1}{2} \times 4 \times 10.5 \times 10.5 = 220.5\,J$$

* K.E of block w.r.to S (ground frame)

$$K = \frac{1}{2} m v^2$$

$$= \frac{1}{2} \times 4 \times 15.5 \times 15.5$$

$$= 480.5\,J$$

$$V = V_{b|T} + V_T$$

$$= 10.5 + 5$$

$$= 15.5$$

* w.r.to observer S' :-

$$W = F S_{b|T}$$

$$= F \times \frac{1}{2} (a_{b|T}) t^2$$

$$= 14 \times \frac{1}{2} \times 3.5 \times 3.5 \times 3 \times 3 = 220.5\,J$$

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* w.r.to observer S :-

$$W = F(S_b)$$

$$= F(S_{b/T} + S_T)$$

$$= 14\left(5\times3 + \frac{1}{2}\times3.5\times3\times3\right)$$

$$W = 430.5\ J$$

$$\boxed{S_T = 5\times3 = 15}$$

→ A running man has half the K.E that a boy of half his mass has, the man speeds up by 1m/s so as to have same K.E as that of boy. Find the original speed of man and boy?

Case-1 :

man	boy
$\frac{K.E}{2}$	K.E
m	$\frac{m}{2}$
v_1	v_2
$\frac{1}{2}mv_1^2 = \frac{K.E}{2}$	$\frac{1}{2}\frac{m}{2}\cdot v_2^2 = K.E$

$$mv_1^2 = \frac{m}{4}v_2^2$$

$$\boxed{2v_1 = v_2}$$

Case-2 :

man	boy
K.E	K.E
m	$\frac{m}{2}$
v_1+1	v_2

$$\frac{1}{2}m(v_1+1)^2 = \frac{1}{2}\frac{m}{2}v_2^2$$

$$(v_1+1)^2 = \frac{1}{2}(2v_1)^2$$

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$$(v_1+1)^2 = 2v_1^2$$

$$v_1^2 + 2v_1 + 1 = 2v_1^2$$

$$v_1^2 - 2v_1 - 1 = 0$$

$$v_1 = \frac{2 \pm \sqrt{4+4}}{2} = \frac{2 \pm 2\sqrt{2}}{2} = 1 \pm \sqrt{2} = 1 + 1.414$$

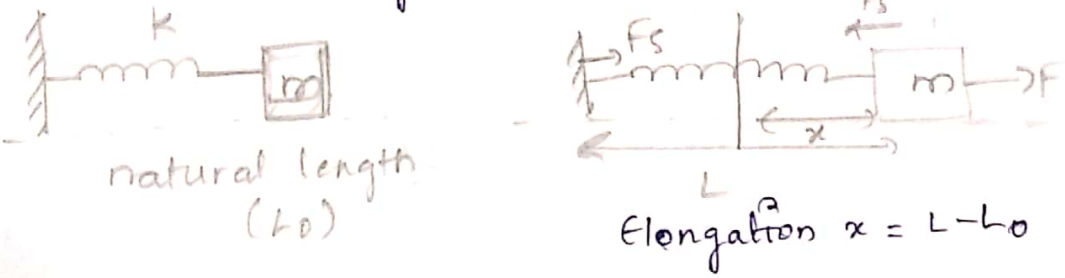
$$v_1 = 2.414 \text{ m/s}$$

$$v_2 = 2(2.414) = 4.828 \text{ m/s}.$$

Work done by spring force :-

spring force $\boxed{F_s = -kx}$; variable

x = elongation / compression in spring.

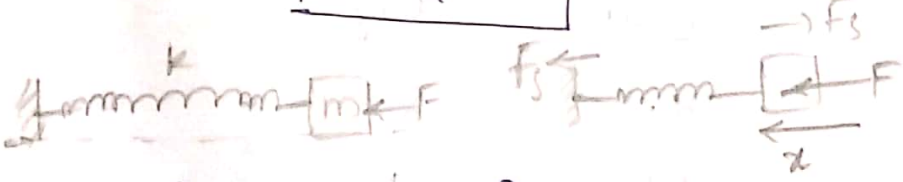


* small work done

$$dw = -|F_s| \cdot dx \quad (\theta = 180^\circ)$$

$$= -kx \cdot dx$$

$$\boxed{w = -\frac{1}{2}kx^2}$$

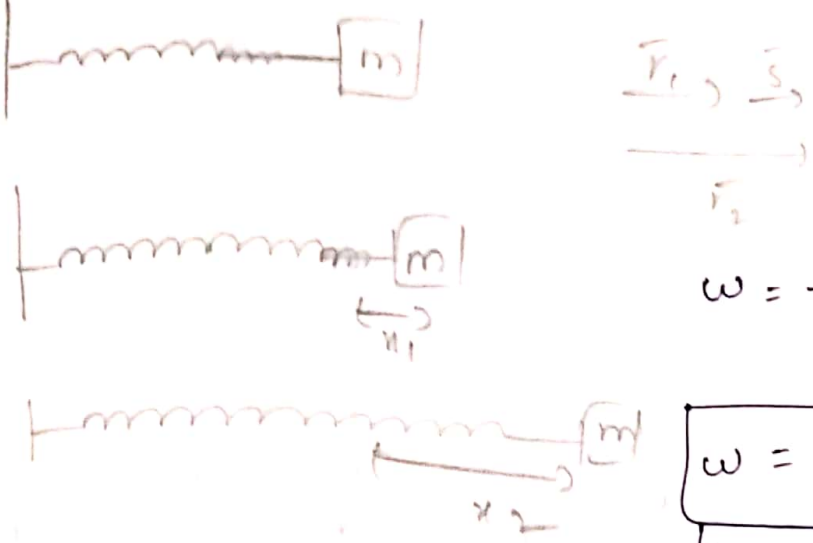


during compression also $w = -\frac{1}{2}kx^2$.

points :

(1) work done by spring force is always (-ve) during elongation / compression from natural length.

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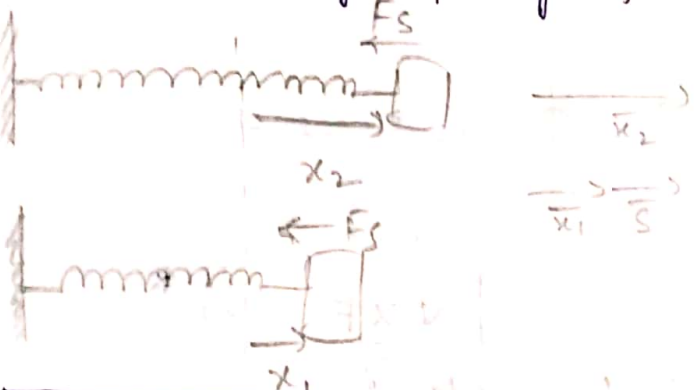


$$W = -\int_{x=x_1}^{x=x_2} kx \cdot dx$$

$$W = -\frac{1}{2} k (x_2^2 - x_1^2)$$

for elongation & compression.

During getting original length, work done by spring force is



$$W = -\int_{x_2}^{x_1} kx \cdot dx$$

$$= -\frac{1}{2} k (x_1^2 - x_2^2)$$

$$W = \frac{1}{2} k (x_2^2 - x_1^2)$$ (+ve)

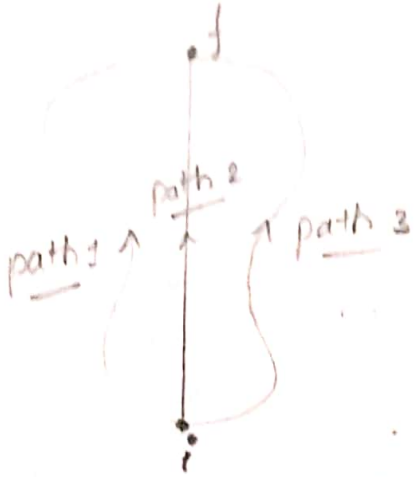
$$W = -\int_{x=x_i}^{x=x_f} kx \cdot dx$$

Conservative and Non-conservative forces:

Conservative forces:-

A force is said to be conservative if workdone by that force in moving the body from initial position to final position is same for all paths & work done by that force in moving the body along a closed path is zero.

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$\omega_{path_1} = \omega_{path_2} = \omega_{path_3}$

$\omega_{closed} = 0$

$$\boxed{\oint_l \bar{F} \cdot d\bar{r} = 0}$$

line integral.

* We have,

$$\oint_l \bar{F} \cdot d\bar{l} = \oint_s curl\, \bar{F} \cdot d\bar{l} \quad \text{(Stoke's theorem)}$$

$$\oint_l \bar{F} \cdot d\bar{l} = 0 \Rightarrow \oint_s curl\, \bar{F} \cdot d\bar{l} = 0$$

$$curl\, \bar{F} = 0$$

$$\boxed{\bar{\nabla} \times \bar{F} = 0} \quad *$$

Here,

$$\bar{\nabla} = i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z} \rightarrow \text{differential operators}$$

↳ partial differentiation.

[curl → rotation]

$$\bar{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$$

$$\bar{\nabla} \times \bar{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

$$= \hat{i}\left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right) - \hat{j}\left(\frac{\partial F_z}{\partial x} - \frac{\partial F_x}{\partial z}\right) + \hat{k}\left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)$$

$$= 0$$

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$\Rightarrow \frac{\partial F_z}{\partial y} = \frac{\partial F_y}{\partial z}, \quad \frac{\partial F_z}{\partial x} = \frac{\partial F_x}{\partial z}, \quad \frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}$.

If $\bar{F} = F_x\uparrow + F_y\uparrow$ [2D]

$$\bar{\nabla} \times \bar{F} = \begin{vmatrix} \uparrow & \uparrow & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & 0 \end{vmatrix} = 0$$

$$= \uparrow\left(-\frac{\partial F_y}{\partial z}\right) - \uparrow\left(-\frac{\partial F_x}{\partial z}\right) + \hat{k}\left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right) = 0$$

$\Rightarrow$ $\boxed{\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}}$

→ $\bar{F} = x^2y\uparrow + xy^3\uparrow$ → Is this conservative or not ?

$F_x = x^2y \qquad F_y = xy^3$

$$\frac{\partial F_x}{\partial y} = \frac{\partial}{\partial y}(x^2y) = x^2 \cdot \frac{\partial y}{\partial y} = x^2$$

$$\frac{\partial F_y}{\partial x} = \frac{\partial}{\partial x}(xy^3) = y^3$$

$\frac{\partial F_x}{\partial y} \neq \frac{\partial F_y}{\partial x}$ $\Rightarrow$ it is not conservative.

Examples :-

① constant vector is a conservative

② $\bar{F} = m\bar{g}$, $\bar{F} = \epsilon \cdot q$

③ gravitational, spring, electrostatic forces are conservative.

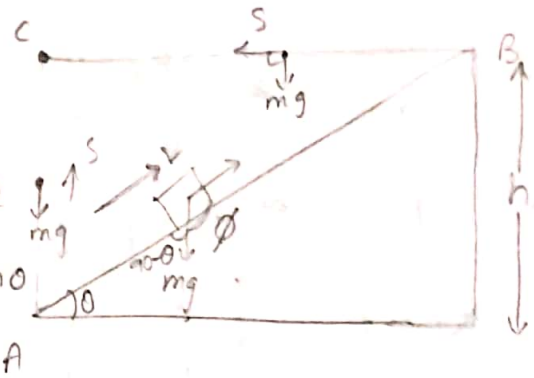
④ Central force



$\bar{v}$ & $\bar{F}$ are perpendicular so $\omega = 0$ $\Rightarrow$ conservative

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→



$\phi + 90 - \theta = 180°$

$\boxed{\phi = 90 + \theta}$

$\sin\theta = \frac{h}{L}$

$h = L\sin\theta$

*work done by gravity along path 1 · i.e,

Along ABC is,

$$W_{ABC} = W_{AB} + W_{BC}$$

$$W_{AB} = mgL\cos\phi \qquad W_{BC} = 0 \ (\theta = 90°)$$

$$= mgL(\cos(90+\theta))$$

$$= -mgL\sin\theta \qquad \Rightarrow \boxed{W_{ABC} = -mgh}$$

$$W_{AB} = -mgh$$

*work done by gravity along path 2 i.e, along AC.

$$\boxed{W_{AC} = -mgh}$$

⇒ clearly $W_{Path_1} = W_{Path_2}$.

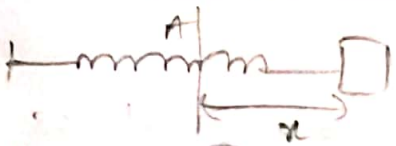
* Work done by gravity along closed path,

$$W_{ABCA} = W_{AB} + W_{BC} + W_{CA}$$

$$= -mgh + 0 + mgh$$

$$\boxed{W_{ABCA} = 0}$$

→



$$W_{AB} = -\frac{1}{2}kx^2$$



$$W_{BA} = \frac{1}{2}kx^2 \qquad \Rightarrow \boxed{W = 0}$$

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Non-Conservative force :-

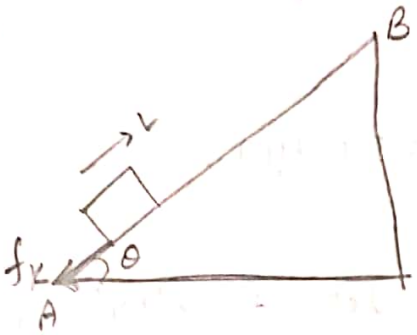
A force is said to be non-conservative if work done by the force in moving the body from initial position to final position is different along different paths. [path dependent force].

Total work done by non-conservative force along closed path is not equal to zero.

→ frictional force, viscous drag etc.

→ energy dissipative force

*



$$W_{AB} = -f_K L = -\mu_K N L$$
$$= -\mu_K mg\cos\theta L$$

$$W_{BA} = -f_K L = -\mu_K N$$
$$= -\mu_K mg\cos\theta L$$

* $W_{Total} = W_{AB} + W_{BA} = -2\mu_K mg\cos\theta L \neq 0$

→ which of the following is/are conservative force(s) ?

(1) $\vec{F} = 2r^3\hat{r}$ (2) $\vec{F} = -\frac{5}{r}\hat{r}$ (3) $\vec{F} = \frac{3(x\hat{\imath}+y\hat{\jmath})}{(x^2+y^2)^{3/2}}$

(4) $\vec{F} = \frac{3(y\hat{\imath}+x\hat{\jmath})}{(x^2+y^2)^{3/2}}$

(1) $\vec{F} = 2r^3\hat{r}$

$W = \int \vec{F}\cdot d\vec{r}$

$= \int 2r^3 \cdot \hat{r}\cdot d\vec{r}$

$= \int 2r^3 . dr = \frac{r^4}{2} + C$

$\hat{r}\cdot d\vec{r} = |\hat{r}|\cdot|d\vec{r}|\cos 0°$

$= dr$

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(2) $\bar{F} = -\frac{5}{r} \cdot \hat{r}$

$$W = \int \bar{F} \cdot d\bar{r} = \int -\frac{5}{r} \cdot \hat{r} \cdot d\bar{r}$$

$$= \int -\frac{5}{r} \cdot dr$$

$$= -5 \ln r + c$$

(3) $\bar{F} = 3\left(\frac{x\hat{i} + y\hat{j}}{(x^2+y^2)^{3/2}}\right)$

$$\int \bar{F} \cdot d\bar{r} = 3\int \frac{(x\hat{i} + y\hat{j})}{(x^2+y^2)^{3/2}} \cdot (dx\hat{i} + dy\hat{j})$$

$$= 3\int \frac{x \cdot dx + y\,dy}{(x^2+y^2)^{3/2}}$$

let $t^2 = x^2 + y^2$

$2t \cdot dt = 2x \cdot dx + 2y \cdot dy$

$t \cdot dt = x \cdot dx + y \cdot dy$

$$= 3\int \frac{t \cdot dt}{t^3}$$

$$= \frac{-3}{t} + c = \frac{-3}{\sqrt{x^2+y^2}} + c$$

(4) $\bar{F} = 3\left(\frac{y\hat{i} + x\hat{j}}{(x^2+y^2)^{3/2}}\right)$

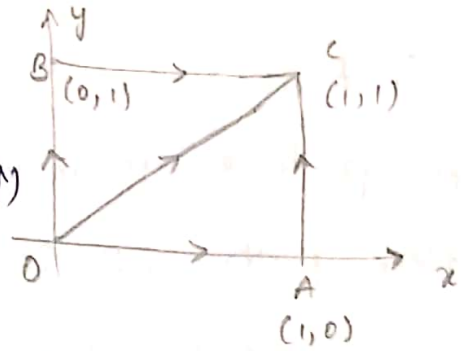
$$\int \bar{F} \cdot d\bar{r} = \int \frac{3(y\hat{i} + x\hat{j})}{(x^2+y^2)^{3/2}} (dx\hat{i} + dy\hat{j})$$

$$= 3\int \frac{y\,dx + x\,dy}{(x^2+y^2)^{3/2}} = 3\int \frac{d(xy)}{(x^2+y^2)^{3/2}}$$

It is not solvable without any additional information.

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→ $\bar{F} = xy\hat{\imath} + xy\hat{\jmath}$ ⟶ Is this a conservative force or not?



$$\int \bar{F} \cdot d\bar{r}$$

$$= \int (xy\hat{\imath} + xy\hat{\jmath}) \cdot (dx\hat{\imath} + dy\hat{\jmath})$$

$$\omega = \int xy\,dx + \int xy\,dy.$$

⟶ it is non conservative force.

* $\omega_{OAC} = \omega_{OA} + \omega_{AC}$

along OA, $y = 0$ & $dy = 0$

$\therefore \omega_{OA} = \int x(0)\cdot dx + \int x(0)\cdot(0) = 0$

Along AC $\Rightarrow x = 1 \Rightarrow dx = 0$

$$\therefore \omega_{AC} = \int (1)\cdot y \cdot (0) + \int_{y=0}^{y=1} (1)(y)\cdot dy$$

$$\boxed{\omega_{AC} = \frac{1}{2}\,J}$$

* $\omega_{OBC} = \omega_{OB} + \omega_{BC}$

along OB; $x = 0$ & $dx = 0$

$\Rightarrow \boxed{\omega_{OB} = 0}$

along BC; $y = 1 \Rightarrow dy = 0$

$$\therefore \omega_{BC} = \int_{x=0}^{x=1} x\cdot dx = \frac{1}{2}\,J$$

$$\boxed{\omega_{BC} = 1/2\,J}$$

along OC:

$x = y$

$dx = dy$

$$\omega_{OC} = \int x^2\cdot dx + \int x^2\cdot dx = 2\int_0^1 x^2\cdot dx = \frac{2}{3}\,J$$

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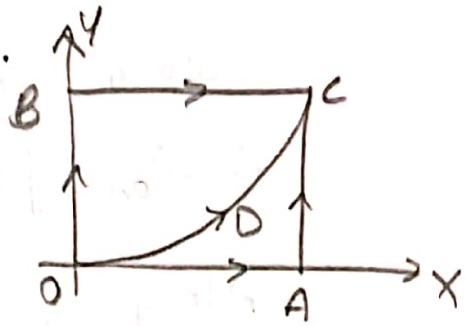
$W_{OAC} \neq W_{OC}$

$\therefore$ F is a non-conservative force.

→ A particle is moved along diff paths as shown. Here path ODC is a parabola $y = ax^2$, & force $\vec{F} = xy\hat{i} + x^2y\hat{j}$ acts on the particle. find —

(a) workdone by the force on the particle along

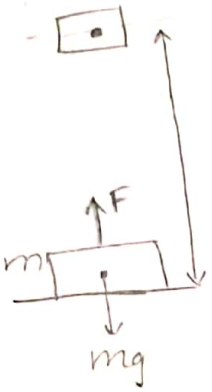
(1) OAC (2) OBC (3) ODC.



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Concept of potential Energy:

* work done by gravity.



$$W = \bar{F} \cdot \bar{S}$$

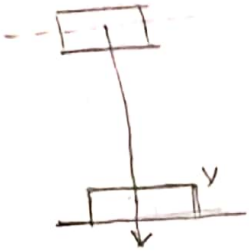
$$= (mg)(h)\cos 180^\circ$$

$$* \boxed{W = -mgh}$$

$$\boxed{W = P.E} \; **$$

$$K.E = \frac{1}{2} m \times 2gh$$

$$\boxed{K.E = mgh}$$



$$K = \frac{1}{2} m v^2$$

$$v = \sqrt{2gh}$$

$$v^2 - u^2 = 2as$$

* P.E → "Energy" possessed by the body by virtue of it's state/position/configuration.
* P.E → position dependent quantity.
* conservative forces → position dependent quantity.
* Relation b/w F_c & P.E (V)

$$\boxed{F = -\frac{dU}{dr}} *$$

$$* \quad \bar{F} = -\hat{i}\frac{\partial U}{\partial x} - \hat{j}\frac{\partial U}{\partial y} - \hat{k}\frac{\partial U}{\partial z}$$

$$\Rightarrow (F_x\hat{i} + F_y\hat{j} + F_z\hat{k}) = -\hat{i}\frac{\partial U}{\partial x} - \hat{j}\left(\frac{\partial U}{\partial y}\right) - k\frac{\partial U}{\partial z}$$

$$\Rightarrow F_x = -\frac{\partial U}{\partial x} \; ; \; F_y$$

$$** \quad \boxed{\bar{F} = -\bar{\nabla} U} * \qquad \bar{\nabla} = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$$

$$* \quad \text{curl } \bar{F} = \bar{\nabla} \times \bar{F}$$

$$\Rightarrow * \quad F = -\frac{dU}{dr}$$

$$\Rightarrow dU = -\bar{F} \cdot d\bar{r}$$

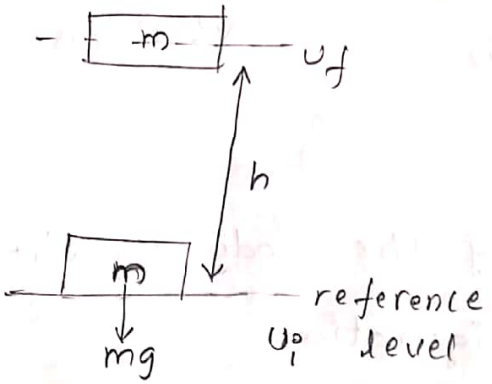
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$$\int_{U_i}^{U_f} dU = -\int_{r_i}^{r_f} F \cdot d\bar{r}$$

$\Rightarrow U_f - U_i$ = - work done by conservative force

$$W_{cons} = -(U_f - U_i)$$

$$\boxed{W_{cons} = U_i - U_f}$$

* 

$W_g = -mgh$

we know that

$W_g = -\Delta U$

$W_g = -(U_f - U_i^\circ)$

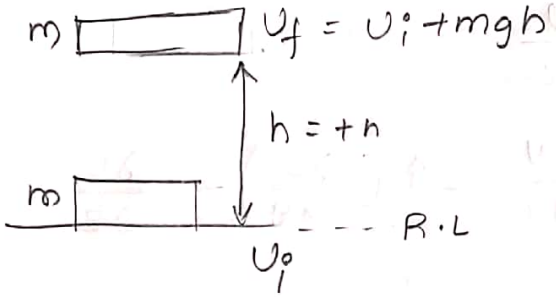
$-mgh = -U_f + U_i^\circ$

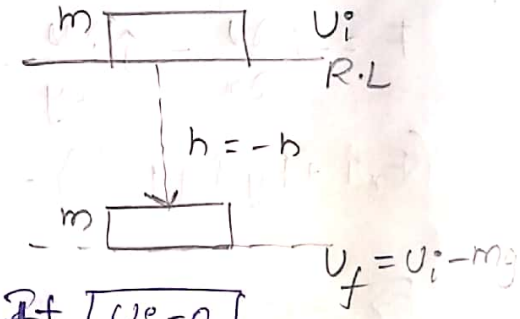
$$\boxed{U_f = U_i^\circ + mgh}$$

* If we consider, $U_i^\circ = 0$ J $\Rightarrow$ $\boxed{U_f = mgh}$ *

$U_i^\circ = 100$ J $\Rightarrow$ $U_f = 100 + mgh$

* $\boxed{U_f - U_i^\circ = mgh}$ *

* 

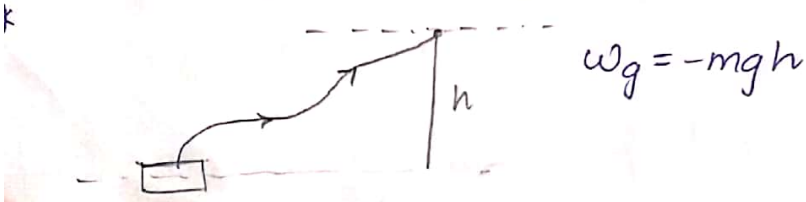


If $\boxed{U_i^\circ = 0}$

then $\boxed{U_f = -mgh}$

P.E may be +ve, -ve (or) zero.

$\boxed{K.E = \frac{1}{2}mv^2}$ $\rightarrow$ +ve.



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→ p.e function as, $U = 2x^2 + 3y^3 + 2z$. Calculate the force acting on the particle at the point (1, 2, 3) m.

$$\bar{F} = -\bar{\nabla} U$$

$$= -\left[\uparrow \frac{\partial U}{\partial x} + \hat{\jmath} \frac{\partial U}{\partial y} + \hat{k} \frac{\partial U}{\partial z}\right]$$

$$\frac{\partial U}{\partial x} = 4x \quad , \quad \frac{\partial U}{\partial y} = 9y^2 \quad , \quad \frac{\partial U}{\partial y} = 2$$

$$\bar{F} = -[4x\uparrow + 9y^2\hat{\jmath} + 2\hat{k}]$$

$$\bar{F}(1,2,3) = -[4(1)\uparrow + 9(2)^2\hat{\jmath} + 2\hat{k}]$$

$$= (-4\uparrow - 36\hat{\jmath} - 2\hat{k})\,N.$$

→ $\bar{F} = 2x\uparrow + 3y^2\hat{\jmath}$ (force in a conservative field). Find potential energy function, if it is zero at the orgin.

$$U_f - U_i = -\int_{(0,0,0)}^{(x,y,z)} \bar{F} \cdot d\bar{r}$$

$$U - 0 = -\int_{(0,0,0)}^{(x,y,z)} (2x\uparrow + 3y^2\hat{\jmath}) \cdot (dx\uparrow + dy\hat{\jmath} + dz\hat{k})$$

$$= -\int_0^x 2x \cdot dx - \int_0^y 3y^2 \cdot dy$$

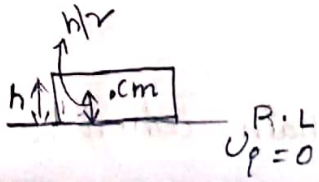
$$= -2\left[\frac{x^2}{2}\right]_0^x - 3\left[\frac{y^3}{3}\right]_0^y$$

$$U = -x^2 - y^3 = -(x^2 + y^3)$$

Calculation of potential energy:

①

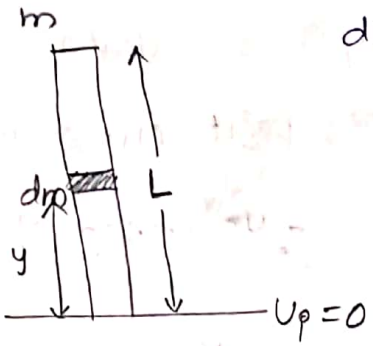
$$U_f = U_i + mgh$$



$$U = mg\frac{h}{2}$$

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(2)



$dU = (dm)\,g\,y$

$U = \int dU$

$U = \int_{y=0}^{L} (dm)\,gy$

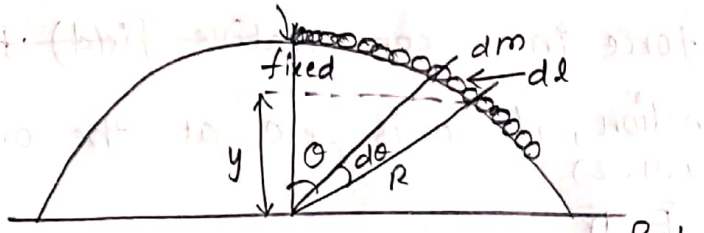
$= \int_{0}^{L} \left(\frac{m}{L}\,dy\right) gy$

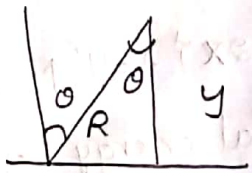
$= \frac{M}{L}\,g\left[\frac{y^2}{2}\right]_0^L = \frac{MgL}{2}$

Π-method :

(COM) $\boxed{U = \frac{mgL}{2}}$

(3) → A chain of length $l < \frac{\pi R}{2}$ is placed on of smooth hemispherical surface. Find the gravitational potential energy stored in chain.





$\cos\theta = \frac{y}{R}$

$y = R\cos\theta$

$dU = (dm)\,gy$

$dU = (\lambda\,dl)\,gy$

$= \lambda\,dl\,g(R\cos\theta)$

$dU = \lambda(R\,d\theta)\,g(R\cos\theta)$

$\int dU = \int \lambda R^2 g(\cos\theta)\cdot d\theta$

$U = \lambda R^2 g \int_{\theta=0}^{\theta=l/R} \cos\theta\cdot d\theta$

$= \lambda R^2 g\,[\sin\theta]_0^{l/R}$

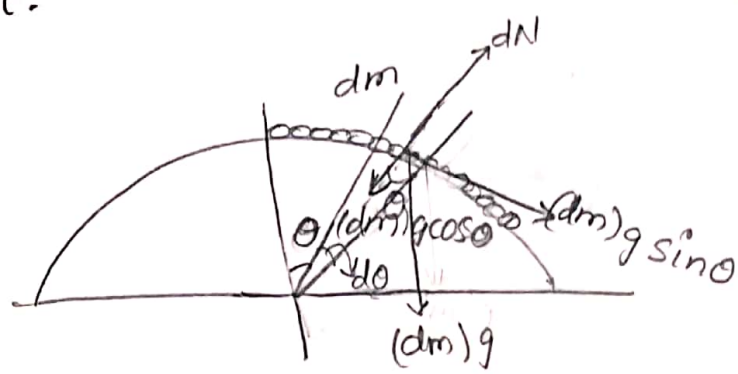
$= \lambda R^2 g \sin(l/R)$

$= \frac{M}{l}\,gR^2\sin(l/R)$

→ Find tangential acceleration of chain when it is

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released from this point.



$$dN = (dm)\, g\cos\theta$$

Tangential force on small element,

$$dF_t = (dm)\, g\sin\theta$$

$$dF_t = \frac{m}{L}(dl)\, g\sin\theta$$

$$dF_t = \frac{m}{L} R \cdot d\theta\, g\sin\theta$$

$$= \frac{m}{L}(R\,d\theta)\, g\sin\theta$$

∴ Total tangential force,

$$F_t = \int dF_t = \frac{mgR}{L}\int_{\theta=0}^{\theta=l/R} \sin\theta \cdot d\theta = \frac{mgR}{l}(-\cos\theta)\Big]_0^{l/R}$$

$$F_t = \frac{mgR}{L}\left[-\cos(l/R) + \cos 0\right]$$

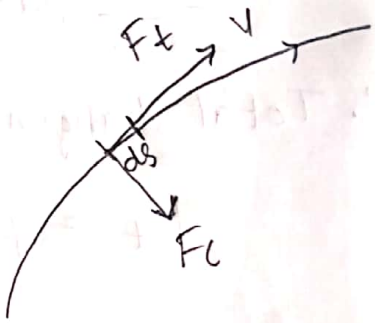
$$\boxed{F_t = \frac{mgR}{l}\left[1 - \cos(l/R)\right]}$$

$$a_t = \frac{F_t}{m}$$

$$\boxed{a_t = \frac{gR}{l}\left(1 - \cos(l/R)\right)}$$

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Work - Energy - Theorem:



* work done by tangential force

$$dw_1 = F_t \cdot ds$$

$$dw_1 = ma_t\, ds$$

$$\Rightarrow dw_1 = m \cdot v \cdot \frac{dv}{ds} \cdot ds$$

$$\Rightarrow dw_1 = mv \cdot dv$$

$$w_1 = \int dw_1$$

$$= m\int_{v_i}^{v_f} v \cdot dv$$

$$= \frac{m}{2}\left(v_f^2 - v_i^2\right)$$

$$= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

* work done by centripetal force

$$w_2 = 0$$

∴ Total work done,

$$w = w_1 + w_2 = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

$$\Rightarrow \boxed{w = \Delta K}$$

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* Finally,

$$\boxed{W_{\text{all force}} = \Delta K}$$

i.e, $$\boxed{W_{\text{conser}} + W_{\text{non-conser}} + W_{\text{ext}} + W_{\text{internal}} = \Delta K}$$

* we have, $W_{\text{conser}} = -\Delta U$

$$\therefore W_{\text{non-cons}} + W_{\text{ext}} + W_{\text{internal}} = \Delta K + \Delta U$$

$$\Rightarrow W_{\text{noncons}} = \Delta(K+U)$$

$$W_{\text{noncons}} = \Delta E$$

Total mechanical energy, $E = K + U$

$$\boxed{W_{\text{noncons}} = E_f - E_i}$$

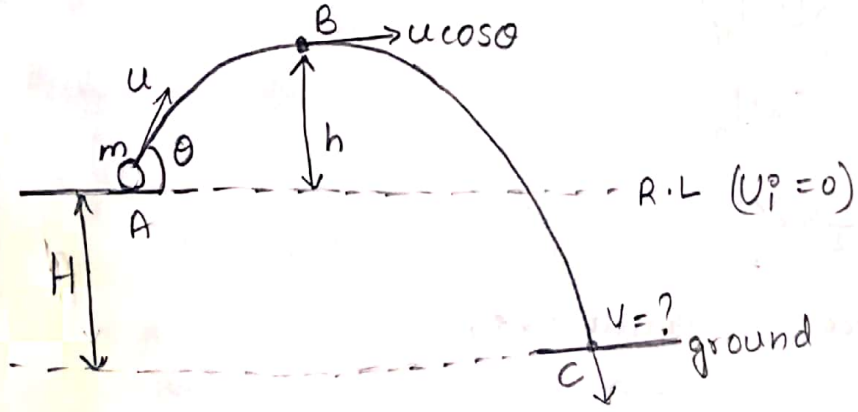
* If $W_{\text{noncons}} = 0$; $W_{\text{ext}} = 0$ then $\Delta E = 0 \Rightarrow E_f = E_i$

$$\Rightarrow \boxed{K_f + U_f = K_i + U_i}$$

"law of conservation of mechanical energy."

Total mechanical energy is of isolated system is constant if internal forces doing work or conservative and external forces do no work.

→



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By Law of conservation of Energy:-

① $E_A = E_B$

$\Rightarrow \frac{1}{2}mu^2 + 0 = \frac{1}{2}mu^2\cos^2\theta + mgh$ — ①

$\frac{1}{2}\cancel{m}u^2 = \frac{1}{2}\cancel{m}u^2\cos^2\theta + \cancel{m}gh$

$u^2 = u^2\cos^2\theta + 2gh$

② $E_A = E_C$

$K_A + U_A = K_C + U_C$

$\frac{1}{2}mu^2 + 0 = \frac{1}{2}mv^2 + mg(-H)$

$\cancel{m}u^2 = \cancel{m}v^2 - 2\cancel{m}gH$

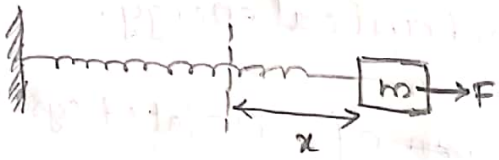
$u^2 = v^2 - 2gH$

$v^2 = u^2 + 2gH$

$$\boxed{v = \sqrt{u^2 + 2gH}}$$

Elastic Potential Energy stored in the spring:

*



$$W_S = -\int_{x_i}^{x_f} Kx\,dx$$

$$= -\frac{K}{2}[x^2]_{x_i}^{x_f}$$

$$= -\frac{1}{2}K[x^2]_0^x$$

$$\boxed{W_S = -\frac{1}{2}Kx^2}$$

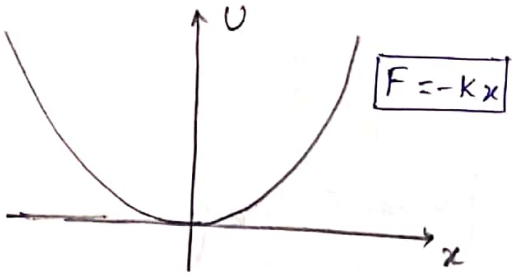
* Spring force is conservative,

$\therefore \quad W_S = -(\Delta U)$

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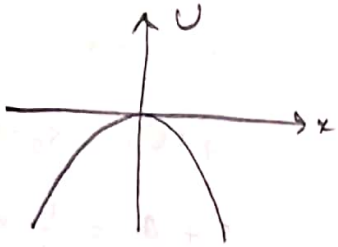
$+\frac{1}{2}kx^2 = +(U_f - U_i)^0$

$\boxed{U = \frac{1}{2}kx^2}$

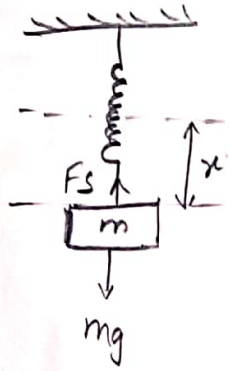


* for $\boxed{F = kx}$ ← conservative force

$$W = \int_{x=0}^{x=x} kx\,dx = \frac{1}{2}kx^2 \Rightarrow U = \frac{1}{2}kx^2$$



Spring - block system:

** 

when block is slowly lowered,

when block is in equilibrium,

$F_{net} = 0$. $a = 0$

$mg = kx$

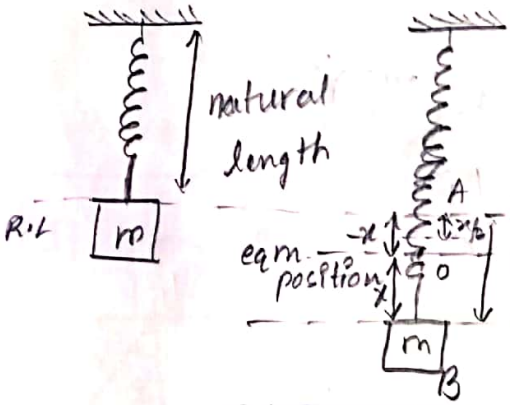
$\boxed{x = \frac{mg}{k}}$

$v \cdot \frac{dv}{dx} = 0$

i.e, v = maximum.

* Maximum elongation in spring:

↳ we use work energy theorem

* 

*when block is suddenly released, we have,

$W_{all} = \Delta K$

$W_{sp} + W_g = K_f - K_i$

$-\frac{1}{2}kx_m^2 + mgx_m = 0$

$mgx_m = \frac{1}{2}kx_m^2$

$\boxed{\frac{2mg}{k} = x_m}$

$\Rightarrow \boxed{x_{max} = 2x}$

* To find max speed of block,

Above $W_g + W_{sp} = \Delta K$

$mgx + (-\frac{1}{2}kx^2) = K_0 - K_A$

$mgx - \frac{1}{2}kx^2 = \frac{1}{2}mv_m^2$

$\frac{m^2g^2}{k} - \frac{1}{2}k\frac{m^2g^2}{k^2} = \frac{1}{2}mv_m^2$

* L.C.E b/w A & B:-

$E_A = E_B$

$K_A + U_A = K_B + U_B$

$0 + 0 = 0 - mgx_m + \frac{1}{2}kx_m^2$

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$$\frac{mg^2}{2k} = \frac{V_m^2}{x}$$

$$x_m = \frac{2mg}{k}$$

$$V_{max} = g\sqrt{\frac{m}{k}}$$

* L.C.E b/w A & B:-

$$E_A = E_0$$

$$K_A + U_A = K_0 + U_0$$

$$0 + 0 = \frac{1}{2}mV_{max}^2 + \left(\frac{1}{2}kx^2 - mgx\right)$$

$$\frac{1}{2}mV_m^2 = mgx - \frac{1}{2}kx^2$$

$$V_{max} = g\sqrt{\frac{m}{k}}$$

* To find velocity at x/2:

L.C.E:

$$0 + 0 = \frac{1}{2}k\left(\frac{x}{2}\right)^2 - mg\left(\frac{x}{2}\right) + \frac{1}{2}mv^2$$

$$\frac{1}{2}mv^2 = mg\frac{x}{2} - \frac{1}{2}\frac{k}{4}(x)^2$$

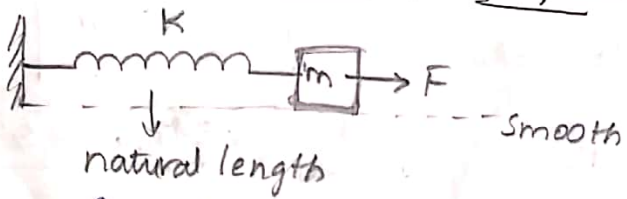
$$mv^2 = mg\left(\frac{mg}{k}\right) - \frac{1}{4}k\left(\frac{mg}{k}\right)^2$$

$$mv^2 = \frac{m^2g^2}{k} - \frac{1}{4}k\frac{m^2g^2}{k^2}$$

$$v^2 = \frac{mg^2}{k}\left(\frac{3}{4}\right)$$

$$v = \frac{g}{2}\sqrt{\frac{3m}{k}}$$

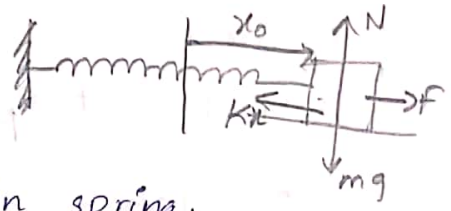
* On smooth horizontal surface:



* elongation in spring when block is at eqm.

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* $F = kx_0 \Rightarrow \boxed{x_0 = \frac{F}{k}}$

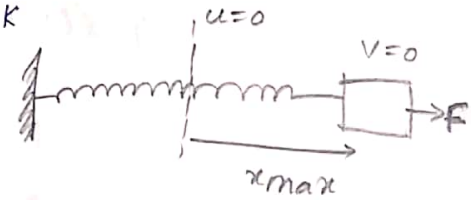


* To find max elongation in spring.

W.E. Theorem,

$$W_F + W_{sp} = \Delta K$$

$$Fx_{max} - \frac{1}{2}kx_{max}^2 = 0$$



$$\boxed{x_{max} = \frac{2F}{k} = 2x_0}$$

* To find max speed of block.

W.E. Theorem,

$$Fx_0 - \frac{1}{2}kx_0^2 = \frac{1}{2}mv_{max}^2$$

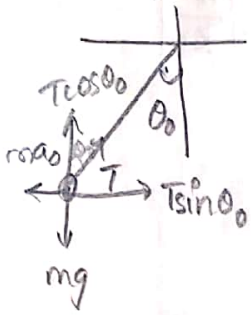
$$\frac{2 \times F \times F}{2 \times k} - \frac{1}{2} k \cdot \frac{F^2}{k^2} = \frac{1}{2} m v_{max}^2$$

$$\frac{2F^2}{k} - \frac{F^2}{k} = m v_m^2$$

$$\frac{F^2}{km} = v_m^2 \Rightarrow v_{max} = F/\sqrt{mk}$$

Work energy Theorem in Non-Inertial Frames

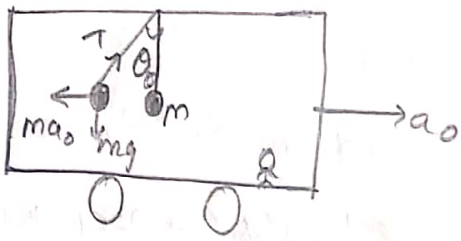
* If pendulum is at eqm,



$F_{net} = 0$

$T\cos\theta_0 = mg$

$T\sin\theta_0 = ma_0$



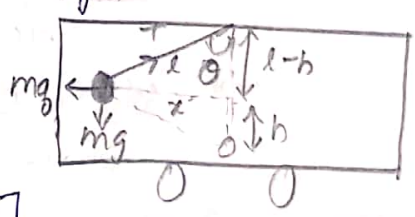
$$\Rightarrow \boxed{\tan\theta_0 = \frac{a_0}{g}} \Rightarrow \boxed{\theta_0 = \tan^{-1}(a_0/g)}$$

* To find max deflection of string:

$$\cos\theta = \frac{l-h}{l}$$

$$h = l(1-\cos\theta)$$

$$\sin\theta = \frac{x}{l} \Rightarrow \boxed{x = l\sin\theta}$$



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W.E. Theorem,

$$W_T + W_g + W_{pseudo} = \Delta K$$

$$0 - mgh + ma_0 x = 0$$

$$g(x(1-\cos\theta)) = a_0 x \sin\theta$$

$$g(1-\cos\theta) = a_0 \sin\theta$$

$$g \cdot 2\sin^2\frac{\theta}{2} = a_0 \cdot 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$$

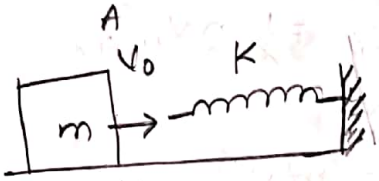
$$g \tan\frac{\theta}{2} = a_0$$

$$\tan\frac{\theta}{2} = \frac{a_0}{g}$$

$$\tan\frac{\theta}{2} = \tan\theta_0$$

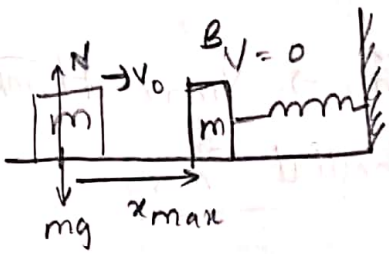
$$\frac{\theta}{2} = \theta_0 \Rightarrow \boxed{\theta = 2\theta_0}$$

→



Smooth

Find max compression in spring



W.E. Theorem,

$$W_N + W_g + W_{sp} = \Delta K$$

$$0 + 0 - \frac{1}{2}kx_m^2 = 0 - \frac{1}{2}mV_0^2$$

$$kx_m^2 = mV_0^2$$

(or)

$$\boxed{x_m = V_0\sqrt{\frac{m}{k}}}$$

→

K.C.E,

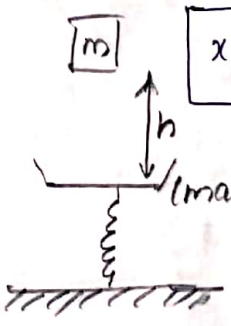
$$K_A + U_A = K_B + U_B$$

$$\frac{1}{2}mV_0^2 + 0 = 0 + \frac{1}{2}kx^2 + 0$$

$$\frac{1}{2}mV_0^2 = \frac{1}{2}kx^2$$

$$\boxed{x = V_0\sqrt{\frac{m}{k}}}$$

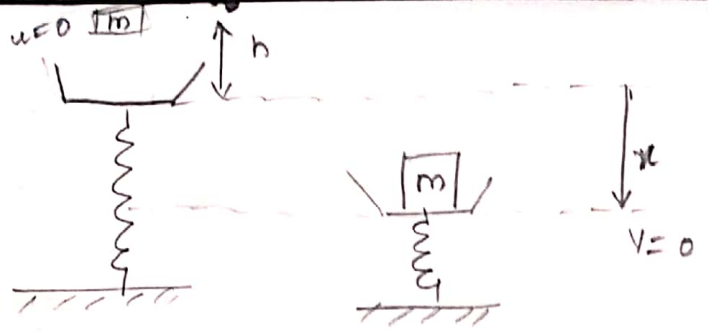
→



block is dropped from certain height 'h' :-

* Max compression in spring:

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$$W_g + W_{sp} = \Delta K$$

$$mg(h+x) - \frac{1}{2}kx^2 = 0$$

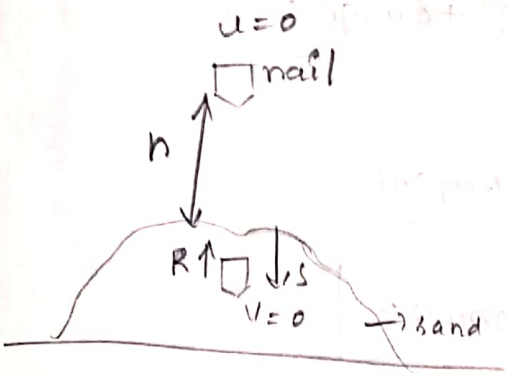
$$mgh + mgx - \frac{1}{2}kx^2 = 0$$

$$x = \frac{2mg \pm \sqrt{4g^2m^2 + 8kmgh}}{2k}$$

$$x = \frac{mg}{k} \pm \frac{\sqrt{m^2g^2 + 2mghk}}{k}$$

$$\therefore \boxed{x = \frac{mg}{k}\left[1 + \sqrt{1 + \frac{hk}{mg}}\right]}$$

→



$$W_g + W_R = \Delta K$$

$$mg(h+s) - Rs = 0$$

$$mgh + mgs = Rs$$

$$mgh = (R-mg)s$$

$$\boxed{s = \frac{mgh}{(R-mg)}}$$

→



Find height 'h' so that lower block just lifts up after hitting with ground.

$$F_{net}$$

$$(m_1+m_2)g = (m_1+m_2)a$$

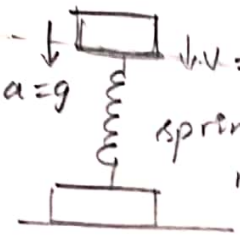
$$\boxed{a = g}$$

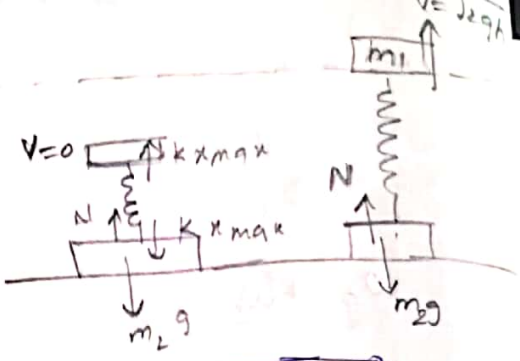
$$m_2g - kx = m_2a$$

$$\boxed{x = 0}$$

there will be no elongation or compression.

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$$N = m_2g$$

$$N = m_2g + Kx_{max}$$

* To find x_{max}:

$$W_g + W_{sp} = \Delta K$$

$$mgx_{max} - \frac{1}{2}kx_{max}^2 = 0 - \frac{1}{2}mv^2$$

$$m_1gx_m - \frac{1}{2}Kx_m^2 = 0 - \frac{1}{2}m(2gh)$$

$$\frac{1}{2}Kx_m^2 - m_1gx_m - \frac{1}{2}mgh = 0$$

$$Kx_m^2 - 2m_1gx_m - 2m_1gh = 0$$

$$x_m = \frac{2m_1g \pm \sqrt{4m_1^2g^2 + 8m_1ghK}}{2K}$$

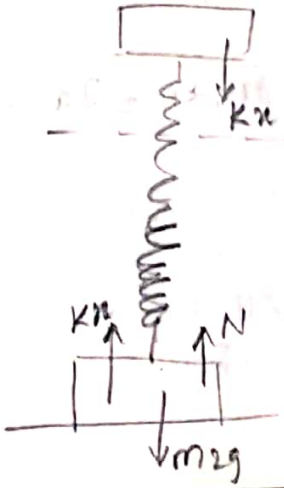
$$= \frac{m_1g}{K} \pm \frac{\sqrt{m_1^2g^2 + 2m_1ghK}}{K}$$

$$\therefore \boxed{x_{max} = \frac{m_1g}{K} + \frac{\sqrt{m_1^2g^2 + 2m_1ghK}}{K}}$$

* acceleration of m_1 at ~~const~~ lowest position:

$$a_1 = \frac{Kx_{max} - m_1g}{m_1}$$

*



$$Kx + N = m_2g$$

$$\boxed{N = m_2g - Kx}$$

To just lift, N = 0

$$\boxed{x = \frac{m_2g}{K}}$$

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* h.C.E,

$$\frac{1}{2}m_1v^2 = m_1gx + \frac{1}{2}kx^2$$

$$\frac{1}{2}m_1v^2 = m_1g\left(\frac{m_2g}{k}\right) + \frac{1}{2}k\left(\frac{m_2g}{k}\right)^2$$

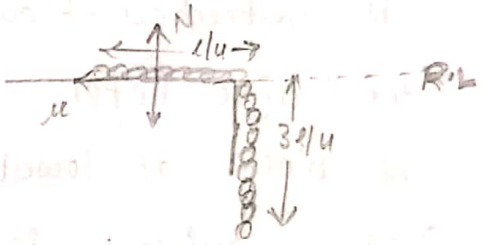
$$\frac{1}{2}m_1v^2 = \frac{m_1m_2g^2}{k} + \frac{1}{2}\frac{m_2^2g^2}{k}$$

$$\frac{1}{2}m_1v^2 = \frac{m_1m_2g^2}{k} + \frac{m_2^2g^2}{2k}$$

$$\frac{1}{2}m_1(2gh) = \left(\frac{2m_1m_2 + m_2^2}{2k}\right)g^2$$

$$\boxed{h = \frac{(2m_1m_2 + m_2^2)g}{2m_1k}}$$

chain is on the vedge of slipping. Find the velocity of chain when it has slipped.

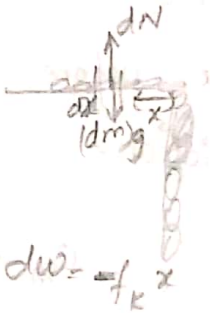


* Here $F = f_l$

$$\frac{3m}{4} = \mu\frac{m}{4}$$

$$\boxed{\mu = 3}$$

* $W_g + W_f + W_N = \Delta K$



$$-(\Delta U) + W_f = \Delta K$$

$$W_f = \Delta K + \Delta U$$

$$W_f = \Delta K + U_f - U_i$$

$$\frac{-3mgl}{32} = \Delta K + U_f - \left[0 - \frac{3m}{4}g\left(\frac{3l}{8}\right)\right]$$

$$\frac{-3mgl}{32} = \Delta K + U_f + \frac{9mgl}{32}$$

$$\frac{-12mgl}{32} = \Delta K + mg\left(-\frac{l}{2}\right)$$

$dW = -f_k x$

$= -(\mu dN)x$

$= -\mu(dm)gx$

$= -\mu\left(\frac{M}{L}\right)dx\,g\,x$

$W = -\frac{\mu Mg}{L}\int_{x=0}^{x=3l/4} x\,dx$

$= -\frac{\mu mg}{L}\left[\frac{(3l)^2}{16}\right]$

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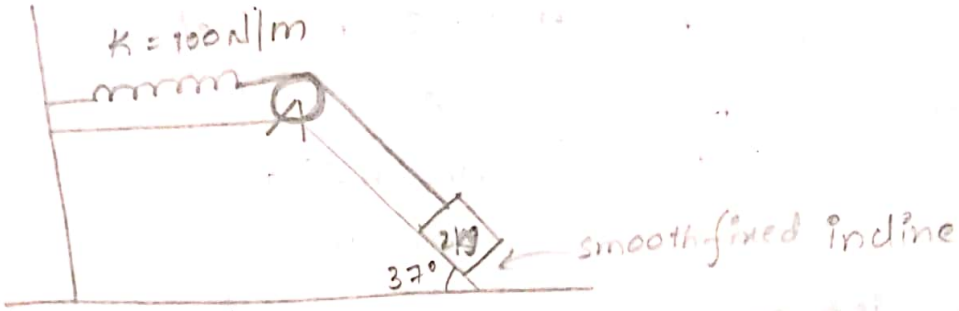
$$\frac{-3mgl}{8} + \frac{mgl}{2} = \Delta K$$

$$\frac{-3mgl + 4mgl}{8} = K_f - K_i^{\nearrow 0}$$

$$\frac{mgl}{8} = \frac{1}{2}mv^2$$

$$v = \frac{\sqrt{gl}}{2}$$

→



① The block is released from rest when the spring is unstreched. How far does the block moves down the incline before coming to rest. what is the acceleration of block at lowest point?

② The experiment is repeated on a rough incline. If the block is observed to move 0.2m down the incline before coming to rest. Calculate the co-efficient of friction.

① $W_g + W_{sp} + W_N^{\nearrow 0} = \Delta K$

$$mg\sin 37° \times x - \frac{1}{2}kx^2 = 0$$

$$mg \times \frac{3}{5} = \frac{1}{2}(10)\,x$$

$$\frac{2 \times 10 \times 3}{5 \times 10 \times 5} = x \Rightarrow \boxed{x = 0.24m}$$

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c) $a = \frac{F_{net}}{m} = \frac{Kx - mg\sin 37^\circ}{m}$

$= \frac{100 \times 0.24 - 20 \times 3/5}{2}$

$= \frac{24 - 12}{2} = 12 - 6 = 6 m/s^2$

* eqm, $Kx_0 = mg\sin 37^\circ$

$x_0 = \frac{20 \times 3/5}{100/5} = \frac{3}{25} = 0.12 m.$

Here velocity of block is max.

W-E Theorem,

$-\frac{1}{2}Kx_0^2 + (mg\sin 37^\circ)x_0 = \frac{1}{2}mv_m^2 - 0$

$\frac{4}{20} \times \frac{3}{5} \times 0.12 - \frac{1}{2}(100)(0.12)^2 = \frac{1}{2}(2)(v_m^2)$

$12 \times 0.12 - \frac{1}{2} \times 12 \times 0.12 = v_m^2$

$\frac{1}{2} \times \overset{6}{12} \times 0.12 = v_m^2$

$v_m = \sqrt{0.72}$

$v_{max} =$ _____

d) $W_g + W_{sp} + W_f + W_N^{\nearrow 0} = \Delta K$

$(mg\sin 37^\circ)s - \frac{1}{2}k(s)^2 - \mu(mg\cos 37^\circ)s = 0$

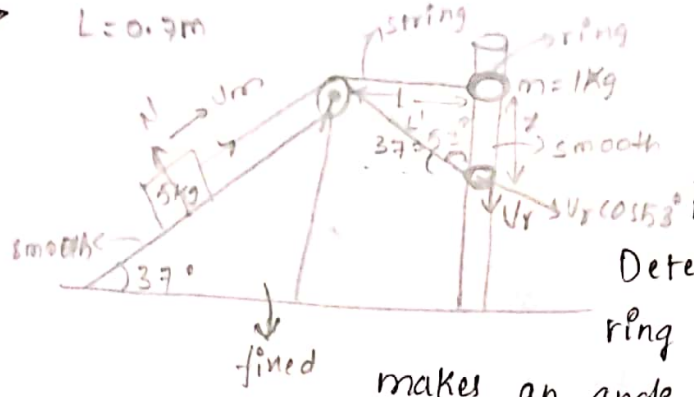
$\frac{4}{20} \times \frac{3}{5} - \frac{1}{2} \times 100 \times \overset{0.1}{0.2} - \mu \times \frac{4}{20} \times \frac{4}{5} = 0$

$12 - 10 = \mu \times 16 \Rightarrow \mu = 2/16 = 1/8.$

$\boxed{\mu = 1/8}$

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→ L = 0.9m



The ring is held in same level with the pulley and released. Determine the velocity of ring & block when string makes an angle 37° with horizontal.

* Constraint eqn,

$$v_m = v_r \cos 53°$$

$$v_m = \frac{3}{5} v_r \quad —(2)$$

* displacement of ring,

$$\tan 53° = \frac{L}{x_r}$$

$$x_r = \frac{3}{4} L = 0.75L$$

* displacement of block,

$$x_b = L' - L$$

$$= \frac{5L}{4} - L$$

$$= \frac{L}{4} = 0.25L$$

* ω-ε Theorem,

$$\omega_g + \omega_T + \omega_N = \Delta K$$

$$(-mg \sin 37°)\, x_b + mg\,(x_r) = \frac{1}{2} m v_r^2 + \frac{1}{2} m v_b^2$$

$$1 \times 10 \times 0.75L - 5 \times 10 \times \frac{3}{5} \times 0.75L$$

$$= 5L - 7.5(3)L$$

$$2.5L\,(1-9) = \frac{1}{2}(1) v_r^2 + \frac{1}{2}(5)\left(\frac{3}{5} v_r\right)^2$$

$$-8(2.5)L = v_r^2 + \frac{9}{5} v_r^2$$

$$-40 \times 0.7 = \frac{13 v_r^2}{5}$$

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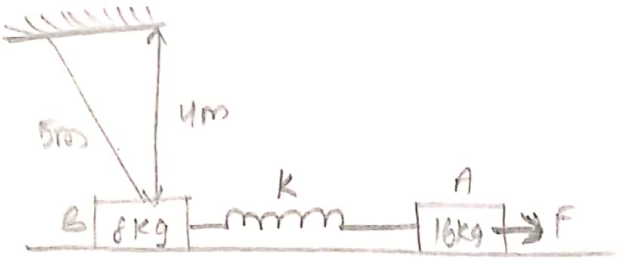
$$-\cancel{5}\times 10\times\frac{3}{\cancel{5}}(0.25)L+1\times 10\times 0.75L=\frac{1}{2}\left(V_r^2+\frac{9}{5}V_r^2\right)$$

$$-0.75L+0.75L=\frac{1}{2}\left(\frac{14}{5}V_r^2\right)$$

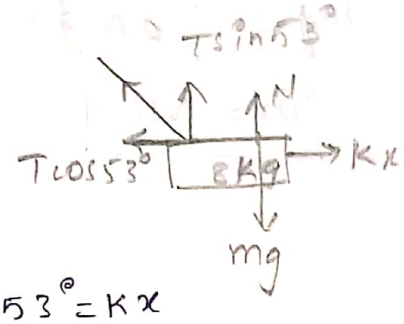
$$\boxed{V_r=0\ m/s}$$

$$\Rightarrow \boxed{V_m=0}$$

Initially spring is at natural length. A const force F is applied on the heavier block. What is the max possible velocity of F so that lighter block doesnot loose contact with the ground.



Let 'x' be the elongation in the spring.



on the 8kg block,

$N+T\sin 53^\circ=mg$

Here $N=0$

$$T\times\frac{4}{5}=8\times 10 \qquad T\cos 53^\circ=Kx$$

$$\boxed{T=100N} \qquad Kx=\frac{80}{100}\times\frac{3}{5}$$

$$\boxed{Kx=60N}$$

on the 16kg mass,

W-E Theorem,

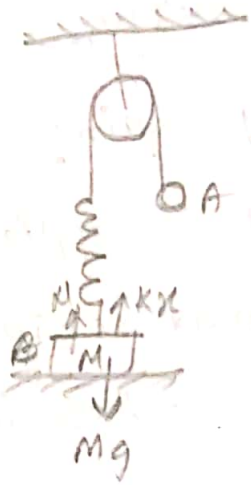
$$Fx-\frac{1}{2}Kx^2=0$$

$$F=\frac{1}{2}Kx=\frac{60}{2}$$

$$\boxed{F=30N}$$

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→ In the figure, a ball A is released from rest when the spring is at its natural length. For the block B of mass M to leave contact with the ground at some stage, the min value of mass of M must be -



$N + Kx = Mg$

here $N = 0$

$Kx = Mg$ —①

W-E Theorem,

$W_g + W_{sp} = \Delta K$

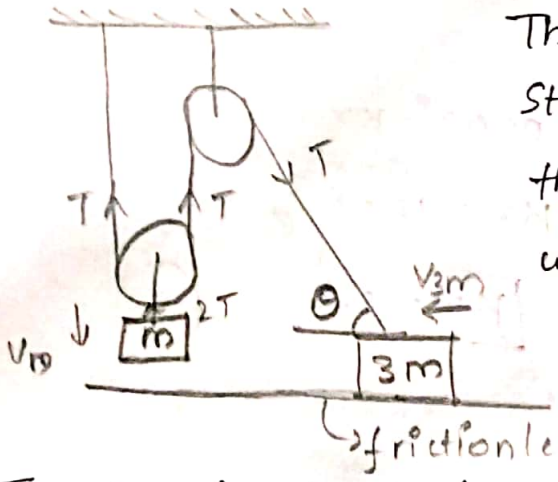
$mgx - \frac{1}{2}Kx = 0$

$mg\cancel{x} = \frac{1}{2}K\cancel{x}$

$mg = \frac{1}{2}Mg$

$\boxed{m = M/2}$

→



The system is ~~released~~ initially held stationary with the inclined thread making an angle 30° with the horizontal. The blocks are now released from rest & allowed to move.

The hanging block falls through a height $\frac{4g}{5}$ before hitting the ground. It is found that the value of θ becomes 60°. Find the speed with which the hanging block hits the floor.

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$$\sum \bar{T} \cdot \bar{v} = 0$$

$$-V_{3m}\cos\theta - 2T v_m = 0$$

$$V_{3m}\cos\theta = 2v_m \longrightarrow (1)$$

$$W_g + W_T^{\nearrow 0} + W_N^{\nearrow 0} = \Delta K$$

$$\Rightarrow mgh = \frac{1}{2}(3m)(V_{3m})^2 + \frac{1}{2}m(v_m)^2$$

$$gh = \frac{3}{2}(V_{3m})^2 + \frac{1}{2}\left(\frac{V_{3m}\cos\theta}{2}\right)^2$$

$$2gh = 3(V_{3m})^2 + \frac{(V_{3m})^2}{4}\cos^2\theta$$

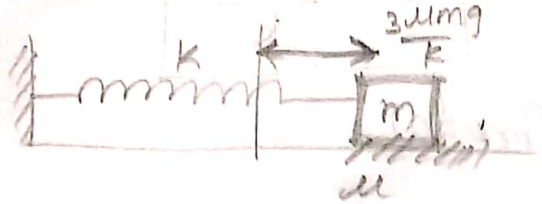
$$\frac{49}{2} = 3(V_{3m})^2 + \frac{(V_{3m})^2}{4} \times \frac{1}{4}$$

$$49 \times 4 = (V_{3m})^2\left(\frac{49}{16}\right)$$

$$V_{3m}^2 = 16 \times 4$$

$$16 \times 4 \times \frac{1}{4} = 4 \times v_m^2$$

$$v_m = 2$$



A spring block system is placed on a rough horizontal surface having su. spring is given initial elongation $\frac{3\mu mg}{k}$ and the is released from rest for the subsequent motion. find

① Initial acc of block

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(2) maximum compression in spring.

(3) maximum speed of block.

(1) $x_i = \frac{3\mu mg}{k} \Rightarrow kx_i = 3\mu mg$

* $a = \frac{F_{net}}{m} = \frac{kx - f}{m} = \frac{3\mu mg - \mu mg}{m}$

$$\boxed{a = 2\mu g}$$

(2) To find max compression,

$$W_{sp} + W_{fk} = \Delta K$$

$$-\frac{1}{2}k(x_f^2 - x_i^2) - \mu mg(x_i + x_f) = 0$$

$$\frac{1}{2}k(x_f + x_i)(x_f - x_i) = \mu mg(x_i + x_f)$$

$$\frac{1}{2}(k)\left(x_f - \frac{3\mu mg}{k}\right) = -\mu mg$$

$$\frac{1}{2}kx_f - \frac{3\mu mg}{2} = -\mu mg$$

$$\frac{1}{2}kx_f = \frac{\mu mg}{2}$$

$$\boxed{x_f = \frac{\mu mg}{k}}$$

(3) $Kx = \mu mg \quad (\because F_{net} = 0)$

$$x = \frac{\mu mg}{k}$$

here velocity is max.

$$W_f + W_{sp} = \Delta K$$

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$\mu mg(x_f - x_i) - \frac{1}{2}k(x_f^2 - x_i^2) = \frac{1}{2}mv^2$

$kx - \frac{1}{2}kx^2 = \frac{1}{2}mv^2$

$\frac{1}{2}kx^2 = \frac{1}{2}mv^2$

$k\left(\frac{\mu mg}{k}\right)^2 = mv^2$

$\frac{\mu^2 m^2 g^2}{k} = mv^2$

$\boxed{v = \mu g\sqrt{\frac{m}{k}}}$

$x_f - x_i$

$= \frac{\mu mg}{k} - \frac{3\mu mg}{k}$

$= \frac{-2\mu mg}{k}$

$x_f^2 - x_i^2$

$= \left(\frac{-2\mu mg}{k}\right)\left(\frac{4\mu mg}{k}\right)$

$= -\frac{8\mu^2 m^2 g^2}{k^2}$

$\mu mg\left(\frac{-2\mu mg}{k}\right) - \frac{1}{2}k\left(\frac{-8\mu^2 m^2 g^2}{k^2}\right) = \frac{1}{2}mv^2$

$-\frac{2\mu^2 m^2 g^2}{k} + \frac{4\mu^2 m^2 g^2}{k} = \frac{1}{2}mv^2$

$\frac{2\mu^2 m^2 g^2}{k} = \frac{1}{2}mv^2$

$\boxed{v = 2\mu g\sqrt{\frac{m}{k}}}$

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Equilibrium:

↳ $F_{net} = 0$

* For a conservative force,

$$F = -\frac{dU}{dr}$$

$$\Rightarrow dU = -F.dr$$

$$\Rightarrow \Delta U = -\int_{r_i}^{r_f} F.dr$$

$$\Rightarrow \boxed{U_f - U_i = -W_F}$$

* only conservative forces are ~~doing~~ doing work, then

$E = \text{const.}$

$$\boxed{K + U = E = \text{const}}$$

we have, $K \geq 0 \Rightarrow \boxed{K_{min} = 0}$ (may be)

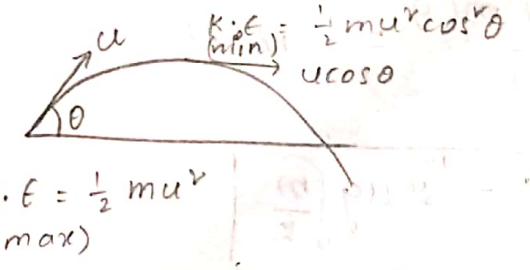
$$\Rightarrow E - U = K \geq 0$$

$$\Rightarrow \boxed{E \geq U}$$

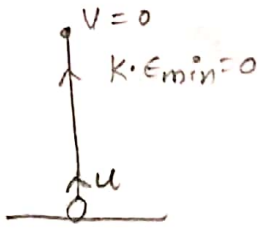
$\boxed{K = U_{max}}$ when $K = 0$

* U_{min} may not be zero.

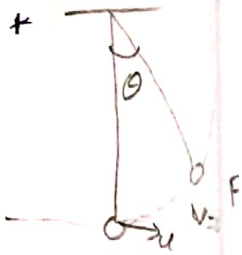
*



*



*



* K.E is positive, can never be -ve.
* min K.E may or may not be zero.
* P.E may be +ve, -ve or zero.
* max P.E is always equal to T.E.
* T.E may be +ve, -ve or zero.
* For equilibrium,

$$F = 0 \Rightarrow \frac{dU}{dr} = 0 \Rightarrow U = \text{const}$$

($\frac{dU}{dr}$ ↳ slope of U/r graph.)

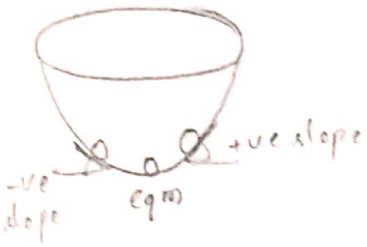
↳ zero
↳ non zero

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Types of Equilibrium :- ← stable equilibrium.
← unstable equilibrium
← neutral equilibrium.

① Stable equilibrium: → $F=0 \Rightarrow \frac{dU}{dr}=0$ & $\frac{d^2U}{dr^2}=+ve$,

$U=U_{min}$.



Unstable equilibrium: → $U=U_{max}$, $\frac{dU}{dr}=0$ & $\frac{d^2U}{dr^2}=-ve$,

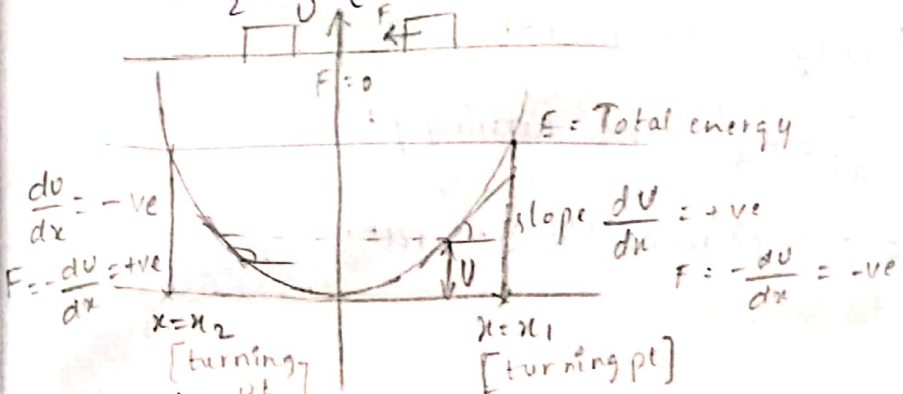
$F=0$.



Neutral equilibrium: → $U=const$, $\frac{dU}{dr}=0$ & $\frac{d^2U}{dr^2}=0$, $F=0$.

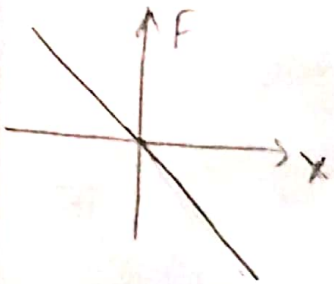


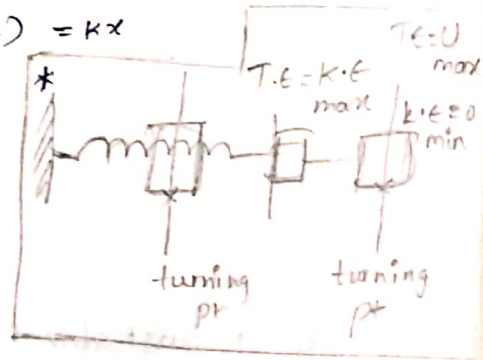
→ For $U=\frac{1}{2}kx^2$ (for 1D-motion)



$\frac{dU}{dx}=\frac{d}{dx}\left(\frac{1}{2}kx^2\right)=\frac{1}{2}k(2x)=kx$

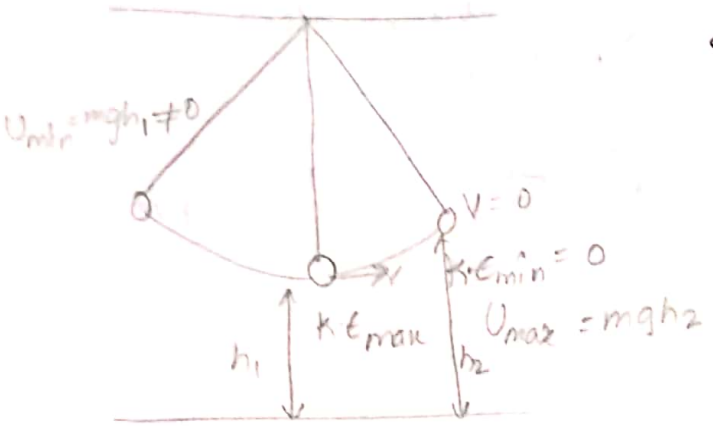
$\boxed{F=kx}$





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*

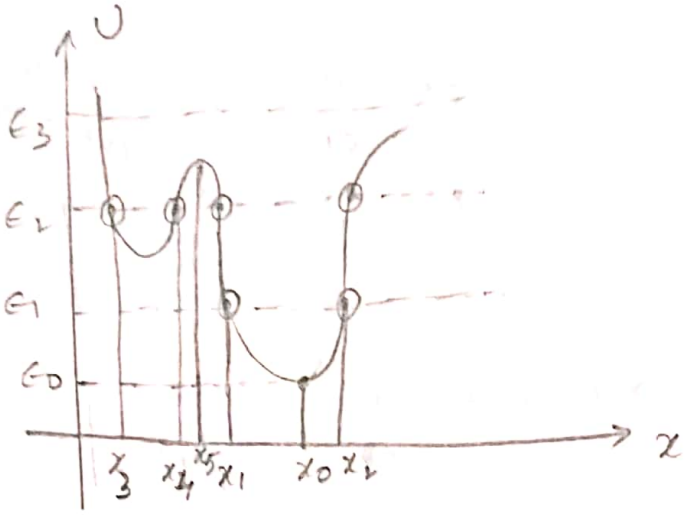


$T.E = const$

$U_{min} + K.E_{max}$

$= U_{max} + K_{min}$ ($K_{min} \to 0$)

*



For $\epsilon_{Total} = \epsilon_1 \longrightarrow x_1, x_2$ are turning pts.

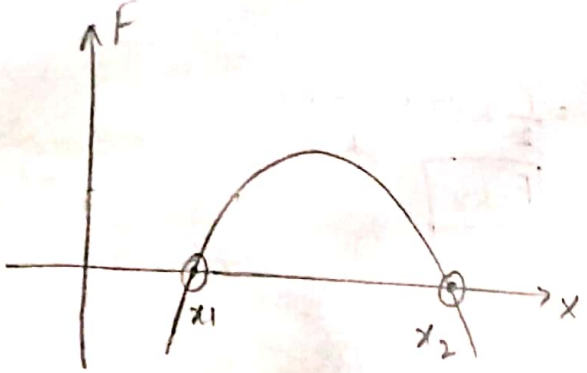
$\epsilon_{Total} = \epsilon_2 \longrightarrow x_3, x_4,$ ~~are~~ x_1, x_2 are turning pts.

at x_5 body is in unstable equilibrium.

$\epsilon_{Total} = \epsilon_3 \longrightarrow$ no turning pt.

$\epsilon_0 \longrightarrow$ permenantly at rest.

*

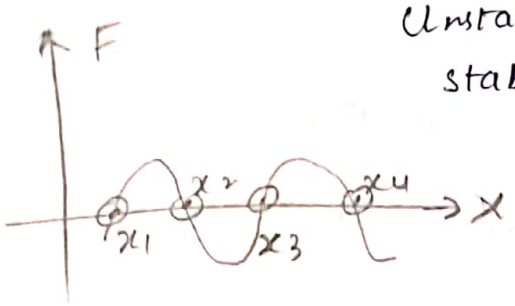


$x_1 \to$ unstable eqm position.

$x_2 \to$ stable eqm position.

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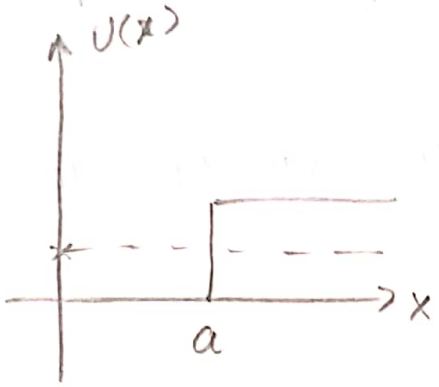
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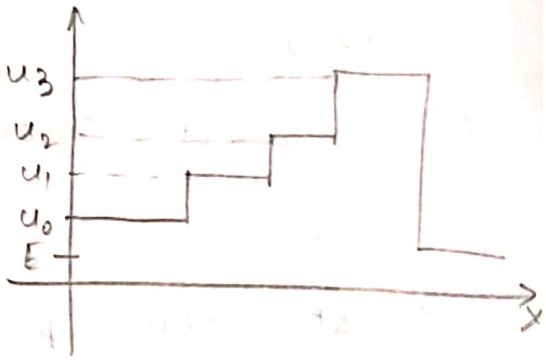
Unstable eqm positions → x_1, x_3

stable eqm positions → x_2, x_4

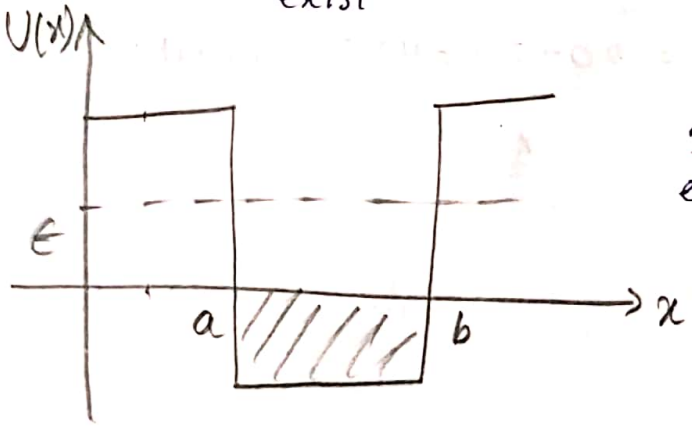
→



For $x > a$, particle doesnot exist



particle doesnot exist.



for particle exist only in the shaded region.

→ A single conversation force $f(x)$ acts on a 1Kg particle that moves along x-axis the P.E $U(x)$ is given by $U(x) = 20 + (x-2)^2$ where x is in (m) at $x = 5m$. particle has a K.E of 20J.

① what is the mechanical energy of system. 49 J

② Make a plot of $U(x)$ as a function of x for $-10 \le x \le 10$

③ The least value of x.

④ The greatest value of x blw which the particle

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can move.

(5) The max K.E of the particle

(6) The value of x at which it occurs.

(7) determine the eqn for $f(x)$ as a function of x.

$\Rightarrow F = -2(x-1)$

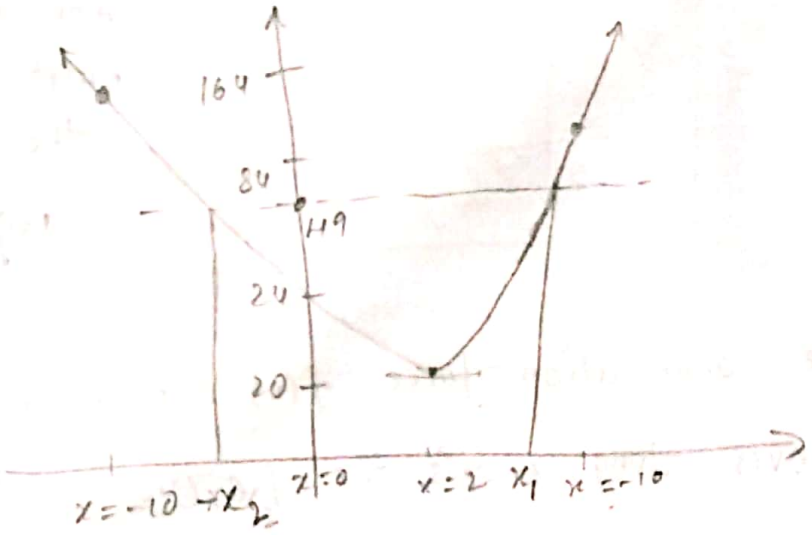
(8) For what finite value of x does $f(x) = 0$. at $x = 2$.

(1) at $x = 5m$, $K.E = 20J$

$$U = 20 + (5-2)^2 = 29\ J$$

$$T.E = 20 + 29 = 49\ J = \text{const.}$$

(2)



$U_{min} + K.E_{max} = T.E$

$20 + K.E_{max} = 49$

$K.E_{max} = 29$

$\frac{1}{2} m v_{max}^2 = 29$

At turning point,

$P.E_{max} = T.E$

$20 + (x-2)^2 = 49$

$(x-2)^2 = 29$

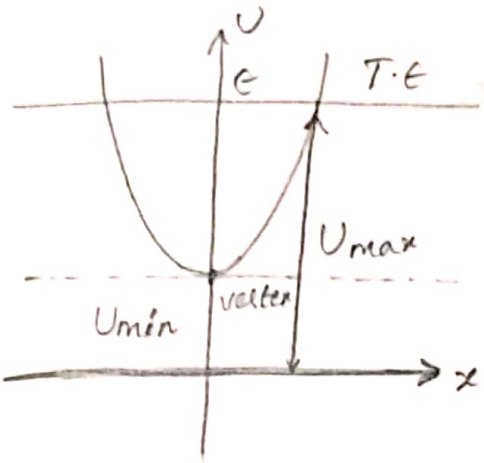
$x = 2 \pm \sqrt{29}$

$F = -\frac{dU}{dx} = -\frac{d}{dx}(20 + (x-2)^2) = -2(x-2)$

$= -2x + 4$



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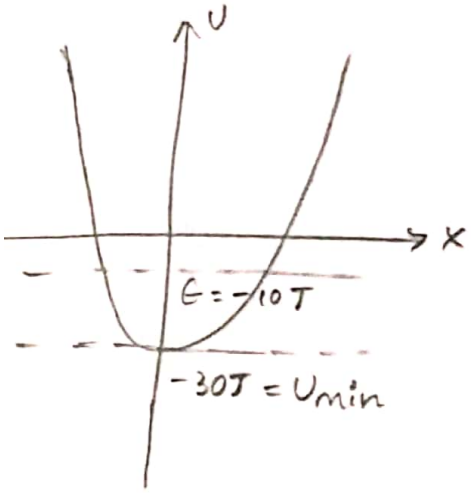
↓ For oscillating system,

$$\boxed{K \cdot \epsilon_{min} = 0}$$

* $U_{max} + K_{min}^{0} = U_{min} + K \cdot \epsilon_{max} = \epsilon$

$\epsilon = U_{max}$ = Energy of particle.

* $U_{max} - U_{min} = K \cdot \epsilon_{max}$ = Energy of oscillation.



$U_{max} = \epsilon = -10\,J$ = energy of particle

oscillating Energy,

$K \cdot \epsilon_{max} = U_{max} - U_{min}$

$= -10 - (-30)$

$= 20\,J$

A particle is moving along x-axis has P.E $(U) = (20 - 20x + 5x^2)$ J. the particle is released at $x = -3m$. Calculate the max value of x.

$U = 2 - 20x + 5x^2$

$K \cdot \epsilon = 0$ at $x = -3m$

$U = 2 - 20(-3) + 5(-3)^2$

$= 2 + 60 + 45$

$\boxed{U = 107\,J}$

$\epsilon = U + K = 107\,J$

energy of oscillator $= \epsilon = 107\,J$

at turning pts, $U = \epsilon$.

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$2 - 20x + 5x^2 = 107$.

$5x^2 - 20x - 105 = 0$

105
× 2
210

$$x = \frac{20 \pm \sqrt{400 + 2100}}{2 \times 5}$$

$$= \frac{20 \pm \sqrt{2500}}{10}$$

$$= \frac{20 \pm 50}{10}$$

$= 2 \pm 5 \Rightarrow x = 7, -3$.

For U_{min}, $\frac{dU}{dx} = 0$

$$\frac{d}{dx}(2 - 20x + 5x^2) = 0$$

$$-20 + 10x = 0$$

$\boxed{x = 2} \Rightarrow$ eqm position.

now, $\frac{d^2U}{dx^2} = \frac{d}{dx}(-20 + 10x) = 10$

$\frac{dU}{dx} = 0$ & $\frac{d^2U}{dx^2} = +ve$

$\Rightarrow$ stable equilibrium.

$U_{min} = 2 - 20(2) + 5(4)$

$= 2 - 40 + 20 = -18$ J.

energy of oscillation $= U_{max} - U_{min}$

$= 107 - (-18)$

$= 125$ J.

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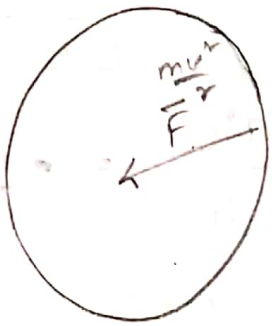
A particle of mass 2 kg is moving under the influence of force which always acts towards the centre and whose P.E is given by $U = 2x^3$ if the body is moving in a circular orbit of radius 5m. find the energy of particle.

$$U = 2r^3$$

$$F = -\frac{dU}{dr}$$

$$F = -6r^2$$

$$|F| = 6r^2$$



$$\frac{mv^2}{r} = 6r^2$$

$$mv^2 = 6r^3$$

$$K = \frac{1}{2}mv^2 = 3r^3$$

$$T = U + K = 5r^3$$

$$T = 5(5)^3 = 5^4 = 625$$

$$\boxed{T = 625\ J}$$

P.E of a particle in a certain field is $U = \frac{a}{r^2} - \frac{b}{r}$ (a, b = +ve const). r is the distance from the centre of the field. Find

① The position of eqm, also state of eqm.

② Find the max magnitude of force of attraction.

③ work done in moving the particle from eqm to ∞.

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given, $U = \frac{a}{r^2} - \frac{b}{r}$

(1) $\frac{dU}{dr} = \frac{-2a}{r^3} + \frac{b}{r^2} = 0$

$\frac{2a}{r^3} = \frac{b}{r^2}$

$\boxed{\frac{2a}{b} = r}$

$\frac{d^2U}{dr^2} = \frac{6a}{r^4} - \frac{2b}{r^3}$

$\left.\frac{d^2U}{dr^2}\right|_{r=\frac{2a}{b}} = \frac{36a}{\frac{16a^4}{b^4}} - \frac{2b}{\frac{8a^3}{b^3}}$

$= \frac{3}{8}\frac{b^4}{a^3} - \frac{b^4}{4a^3}$

$= \frac{b^4}{8a^3} = +ve$

∴ it is stable eqm.

$U_{min} = \frac{a}{\frac{4a^2}{b^2}} - \frac{b}{\frac{2a}{b}} = \frac{b^2}{4a} - \frac{b^2}{2a}$

$= -\frac{b^2}{4a}$

(2) $F = -\frac{dU}{dr} = \frac{2a}{r^3} - \frac{b}{r^2}$

now $F = F_{max}$

$\frac{dF}{dr} = 0 \qquad -\frac{6a}{r^4} + \frac{2b}{r^3} = 0$

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$$\frac{6a^3}{r^4} = \frac{2b}{r^3} \Rightarrow r = \frac{3a}{b}$$

$$F_{max} = \frac{2A}{\frac{27a^3r}{b^3}} - \frac{b}{\frac{9ar}{b^2}}$$

$$= \frac{2b^3}{27a^r} - \frac{b^3}{9ar}$$

$$= \frac{2b^3 - 3b^3}{27a^r}$$

$$\boxed{F_{max} = \frac{-b^3}{27a^r}}$$

* If F = -ve, force is attractive.

* If F = +ve, force is repulsive.

③ $dw = \bar{F} \cdot d\bar{r}$

$= F_{ext} \cdot dr$

$= -F_{field} \cdot dr$

$$dw = -\left(\frac{2a}{r^3} - \frac{b}{r^2}\right) \cdot dr$$

$$dw = \left(\frac{b}{r^2} - \frac{2a}{r^3}\right) \cdot dr$$

$$w = \int_{r=\frac{2a}{b}}^{r=\infty} \frac{b}{r^2} - \frac{2a}{r^3} \cdot dr$$

$$= \int \frac{b}{1} r^{-2} \cdot dr - \int \frac{2a}{1} r^{-3} \cdot dr$$

$$= \frac{b[r^{-1}]_{\frac{2a}{b}}^{\infty}}{-1} + \frac{2a[r^{-2}]_{\frac{2a}{b}}^{\infty}}{-2}$$

$$= -b\left[\frac{1}{r}\right]_{\frac{2a}{b}}^{\infty} + 2a\left[\frac{1}{r^2}\right]_{\frac{2a}{b}}^{\infty}$$

$$w_{ext} = -w_{conserv}$$

$w_{ext} = U_f - U_i$ → at eqm

↳ P.E at ∞ [0]

$$= 0 - \left[-\frac{b^2}{4a}\right]$$

$$\boxed{w_{ext} = \frac{b^2}{4a}}$$

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$$= -b\left[0 - \frac{b}{2a}\right] + a\left[0 - \frac{b^2}{4a^2}\right]$$

$$= \frac{b^2}{2a} - \frac{2ab^2}{4a^2} = \frac{b^2}{2a} - \frac{b^2}{2a} = 0 \cdot \frac{b^2}{4a}.$$

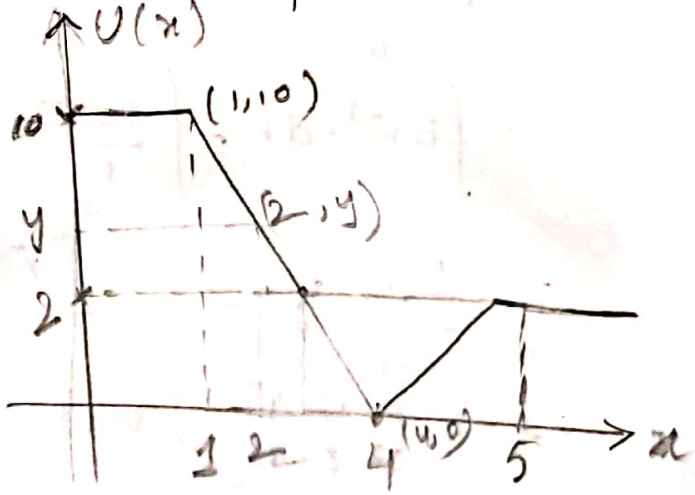
→ A particle is free to move along x-axis has P.E, $U(x) = k\left[1 - e^{-x^2}\right]$ for $-\infty \le x \le \infty$ & K is +ve const then.

(a) At point away from the origin then particle is in unstable equilibrium.

(b) For any finite non-zero value of x there is a force directed away from the origin.

(c) If its total mechanical energy is $\frac{K}{2}$ it has min K.E at the origin & for small displacements from $x = 0$ then motion is simple harmonic.

→ A body with mass 2kg moves in 1 Direction in the presence of force which is decribed by the P.E graph. If the body is released from rest at $x = 2$ m then its speed when crossing $x = 5$ m is ——.



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$$\frac{10-y}{1-2} = \frac{y-0}{2-4}$$

$$\frac{10-y}{+1} = \frac{y}{+2}$$

$$20 = 3y \Rightarrow \boxed{y = \frac{20}{3}}$$

$\therefore$ $T.E = \frac{20}{3}$, $P.E = 2$

$$K.E = \frac{20}{3} - 2 = \frac{14}{3}$$

$$\frac{1}{2}mv^2 = \frac{14}{3}$$

$$\boxed{v = \sqrt{\frac{14}{3}}\ m/s}$$

Power:

Rate of doing work is called "power".

Power $\begin{cases} \text{Avg power} \\ \text{Instantaneous power} \end{cases}$

* Avg power:

$$P_{avg} = \frac{\Delta W}{\Delta t} = \frac{W}{t}$$

ΔW = net work done in a time 'Δt'.

* Instantaneous power:

$$P_{inst} = \lim_{\Delta t \to 0} \frac{\Delta W}{\Delta t} = \frac{dW}{dt}$$

* $\boxed{W_{net} = \Delta K}$ *

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points :

① $P = \frac{\omega}{t}$

* It is a scalar.

* S.I unit ⟶ J/s = watt

1W = 1 J/s

* C.G.S unit ⟶ 1erg/sec

* 1 Hp = 746ω .

② Relation b/w P_{avg} & P_{inst} :

* $P_{inst} = \frac{d\omega}{dt} = \frac{d}{dt}(\vec{F}\cdot\vec{s}) = \vec{F}\cdot\frac{d\vec{s}}{dt} = \vec{F}\cdot\vec{v} = FV\cos\theta$

$$\boxed{P_{inst} = \vec{F}\cdot\vec{v} = FV\cos\theta}$$

* $P_{avg} = \frac{\Delta\omega}{\Delta t} = \frac{\Delta K}{\Delta t} = \frac{\frac{1}{2}mv^2}{\Delta t}$ [if body starts from rest and move with const acceleration]

$= \frac{1}{2}mv\left(\frac{v}{\Delta t}\right)$

$= \frac{1}{2}mav$

$P_{avg} = \frac{1}{2}\vec{F}\cdot\vec{v}$

$v = u + at$

$v = 0 + a\Delta t$

$\Rightarrow \boxed{a = \frac{v}{\Delta t}}$

$$\boxed{P_{avg} = \frac{1}{2}P_{inst}}$$

* $P_{inst} = \frac{d\omega}{dt}$

$d\omega = P_{inst}\cdot dt$

$\Delta\omega = \int_{t_1}^{t_2} P_{inst}\cdot dt$

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* $P_{avg} = \frac{\Delta W}{\Delta t} = \frac{\int_{t_1}^{t_2} P_{ins} \cdot dt}{\int_{t_1}^{t_2} dt}$

) $P = \frac{dW}{dt}$ = slope of tangent of W-t graph gives "instantaneous power".

$P_{avg} = \frac{\Delta W}{\Delta t}$ = slope of secant of W-t graph gives "avg power".

)

$dW = P\,dt$

$\Delta W = \int_{t_1}^{t_2} P \cdot dt = \Delta K$ = area of P-t graph gives.

net work done / change in K.E

Applications:

$P =$ const, $f(t)$, $f(x)$, $f(v)$, W

Eg: ① $P = 5W$

② $P = 2t^2 + 3t + 5$

③ $P = kW$

④ $P = 4x + 3$

Ans: ① $\Delta W = \int_{t_1}^{t_2} P \cdot dt$

If P = const

$\Delta W = P \int_{t_1}^{t_2} dt = P(t_2 - t_1)$

② $\Delta W = \int_{t_1}^{t_2} P \cdot dt = \int_{t_1}^{t_2} (2t^2 + 3t + 5) \cdot dt$

$\Delta W = \left[\frac{2}{3}t^3 + \frac{3}{2}t^2 + 5t\right]_{t_1}^{t_2}$.

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(3) $P = (4x+3)$

$FV = 4x + 3$

$$mv^2 \cdot \frac{dv}{dx} = 4x + 3$$

$$\int_{v_1}^{v_2} mv^2 \cdot dv = \int_{x_1}^{x_2} (4x+3) \cdot dx$$

$$\frac{m}{3}\left[v^3\right]_{v_1}^{v_2} = \left[4x^2 + 3x\right]_{x_1}^{x_2}$$

$$\Delta W = \int P \cdot dt = \int_{t_1}^{t_2} (4x+3) \cdot dt$$

(4) $P = kv$

$F \cdot v = k \cdot v$

$$m \cdot \frac{dv}{dt} = k$$

$$m \cdot dv = k \cdot dt$$

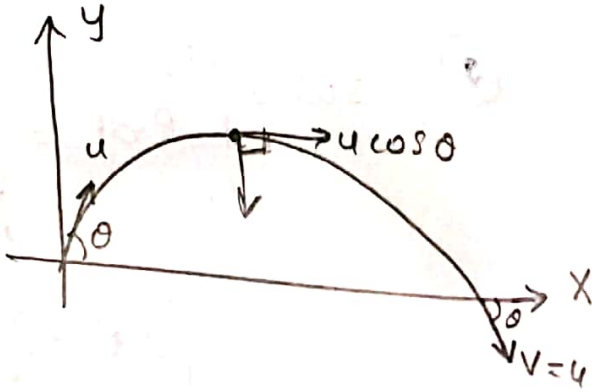
$$\int_{v_1}^{v_2} dv = \frac{k}{m} \int_{t_1}^{t_2} dt$$

$$v_2 - v_1 = \frac{k}{m}(t_2 - t_1)$$

Applications:

(1) Find (a) mean power

(b) ainst power



(a) mean power:

$$\langle P \rangle = \frac{\Delta K}{\Delta t}$$

1) b/w pt of projection & highest pt.

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$$\langle P \rangle = \frac{K_f - K_i}{t_a} = \frac{\frac{1}{2}mu^2\cos^2\theta - \frac{1}{2}mu^2}{\frac{u\sin\theta}{g}}$$

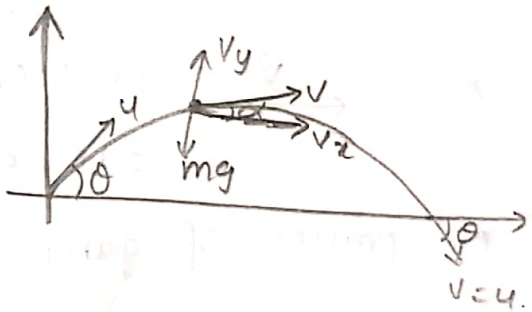
$$\langle P \rangle = \frac{-\frac{1}{2}mu^2\sin^2\theta}{\frac{u\sin\theta}{g}}$$

$$\boxed{\langle P \rangle = -\frac{1}{2}mug\sin\theta}$$

2) b/w pt of projection & point of landing.

$$\langle P \rangle = \frac{\frac{1}{2}mu^2 - \frac{1}{2}mu^2}{\frac{2u\sin\theta}{g}} = 0.$$

(b) inst power :



* $\bar{F} = mg(-\hat{j}) = -mg\hat{j}$

$\bar{V} = V_x\hat{i} + V_y\hat{j}$

$\bar{V} = u\cos\theta\hat{i} + (u\sin\theta - gt)\hat{j}$

* $P_{inst} = \bar{F}\cdot\bar{V}$

$= (-mg\hat{j})\cdot(u\cos\theta\hat{i} + (u\sin\theta - gt)\hat{j})$

* $\boxed{P_{inst} = -mg(u\sin\theta - gt)}$ → $\boxed{P_{inst} = 0 \text{ when } t = \frac{u\sin\theta}{g}}$

P_{inst} is zero when force and velocity are ⊥ler [$\because \cos 90^\circ = 0$].

* $$\langle P \rangle = \frac{\int_{t_1}^{t_2} P_{inst}\cdot dt}{\int_{t_1}^{t_2} dt} = \frac{-mg\int(u\sin\theta - gt)\cdot dt}{t_2 - t_1}$$

$$= \frac{-mg\left(u\sin\theta\cdot t - gt^2\right]_{t=0}^{T}}{t_2 - t_1}$$

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$$= \frac{-mg\cancel{t}(u\sin\theta - g\cancel{t})}{\cancel{t}}$$

$$= \frac{-mg\cancel{(u\sin\theta)}\left(u\sin\theta - g\left(\frac{u\sin\theta}{g}\right)\right)}{\cancel{(u\sin\theta)}}$$

$$= -mg(u\sin\theta - u\sin\theta)$$

$$= 0.$$

** Machine-Gun problems:-

n — no. of bullets fired per sec $= \frac{N}{t}$ *

m — mass of each bullet.

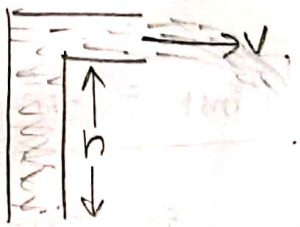
v — speed of bullet while coming out of gun.

* power of gun,

$$P = \frac{\Delta K}{\Delta t} = \frac{\left(\frac{1}{2}mv^2\right)N}{t} = \frac{1}{2}mnv^2.$$

** power of a motor:-

$$P = \frac{\Delta w}{\Delta t}$$



$$P = \frac{mgh + \frac{1}{2}mv^2}{t}$$

$$P = \frac{m}{t}\left(gh + \frac{1}{2}v^2\right)$$

m = mass of water draining in t-sec

$$\boxed{P = \frac{dm}{dt}\left(gh + \frac{v^2}{2}\right)}$$

* Efficiency $\eta = \frac{P_{output}}{P_{input}}$ $\quad \%\ \eta = \frac{P_o}{P_i} \times 100$

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* Crane :-

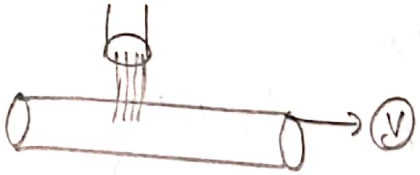
lift load with const speed v.

$$P = F \cdot V$$

$$\boxed{P = (mg)\, v}^{*}$$

$$P = \frac{w}{t} = \frac{mgh}{t} = mgv \rightarrow \text{out put power.}$$

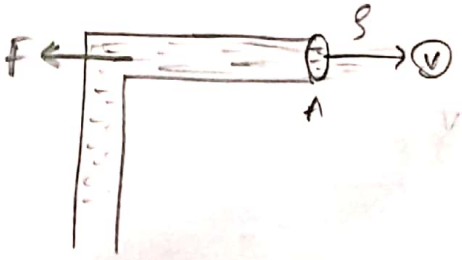
* conveyer belt :-



extra force required to move the belt at const speed, $F = \frac{v\,dm}{dt}$.

extra power required to move the belt at const speed, $P = FV = \frac{v^2 \cdot dm}{dt}$.

* jet of water coming out of pipe :-



* mass of water coming out of the pipe per second,

$$\frac{dm}{dt} = \frac{A\,dl}{dt}\rho = AV\rho,$$

* Force on the water jet by the pipe, $F = v \cdot \frac{dm}{dt} = AV^2\rho$

* power of motor, $P = AV^3\rho$.

** If volume rate is increased by n-times, i.e.,

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Volume rate,

$$\frac{dV}{dt} = \frac{A\,dl}{dt} = AV$$

$$A_1V_1 = n(AV)$$

$$\boxed{V_1 = nV}$$

* Force on the jet of water,

$$F_1 = A_1(nV)^2 \rho = n^2AV^2\rho$$

$$\boxed{F_1 = n^2F}$$

*

* power of ~~the~~ ~~motor~~,

$$\boxed{P_1 = n^3P}$$

$$P_1 = F_1V_1$$
$$= (n^2F)nV$$
$$= n^3FV = n^3P$$

Position & velocity of vehicle as a function of time

* For const power (P):

i.e, P = const

we have, $P = F \cdot V$

$$P = maV$$
$$= m\left(\frac{dV}{dt}\right).V$$

$$\int_{V_1}^{V_2} mV\,dV = \int_{t_1}^{t_2} P\,dt$$

$$\frac{m}{2}(V^2)\Big]_{V_1}^{V_2} = P(t_2 - t_1)$$

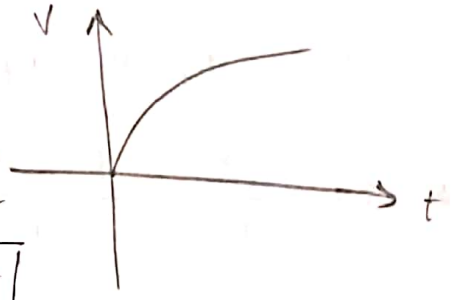
here $V_1 = 0$ & $t_1 = 0$,

$$\frac{m}{2}(V_2)^2 = Pt_2$$

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$$v = \left(\frac{2pt}{m}\right)^{1/2}$$

$\Rightarrow v \propto t^{1/2}$.



$v = kt^{1/2}$

$v^2 = kt$

$\frac{dv}{dt} = \frac{1}{2\sqrt{t}}$ $t \uparrow \Rightarrow v \downarrow$

* we have,

$$v = \left(\frac{2pt}{m}\right)^{1/2}$$

$$\frac{dx}{dt} = \left(\frac{2pt}{m}\right)^{1/2}$$

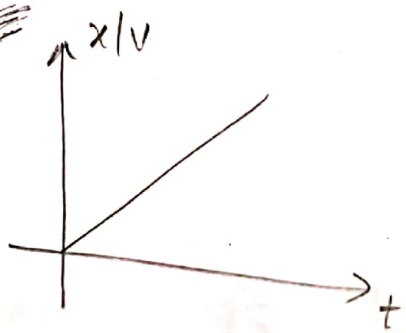
$$\int_{x=0}^{x=x} dx = \left(\frac{2p}{m}\right)^{1/2} \int_{t=0}^{t=t} \sqrt{t}\, dt$$

$$x = \left(\frac{2p}{m}\right)^{1/2} \frac{t^{3/2}}{3/2}$$

$$x = \frac{2}{3}\left(\frac{2p}{m}\right)^{1/2} t^{3/2}$$

$$\boxed{x = \left(\frac{8p}{9m}\right)^{1/2} t^{3/2}}$$

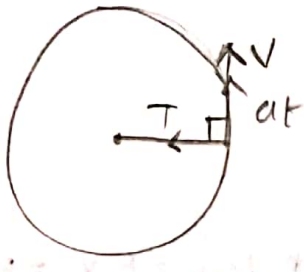
$\boxed{x \propto t^{3/2}}$ *



$\frac{x}{v} = \frac{t^{3/2}}{t^{1/2}} = t$.

**

* $p = \vec{T} \cdot \vec{v} = Tv\cos\theta = 0$.



* $v = krt$

$a_t = \frac{dv}{dt} = kr$.

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$F_t = ma_t = mkr$

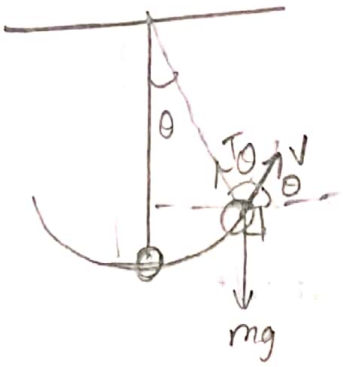
$P_t = \vec{F_t} \cdot \vec{v} = F_t v = (mkr)(krt) = mk^2 r^2 t$

* $a_c = \frac{v^2}{r} = \frac{k^2 r^2 t^2}{r} = k^2 r t^2$

$F_c = ma_c = mk^2 r t^2$

$P_c = \vec{F_c} \cdot \vec{v} = 0$

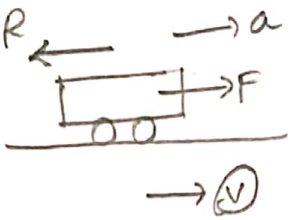
*



$P_g = mgv\cos(90+\theta)$

$= -mgv\sin\theta$

**



$F - R = ma$

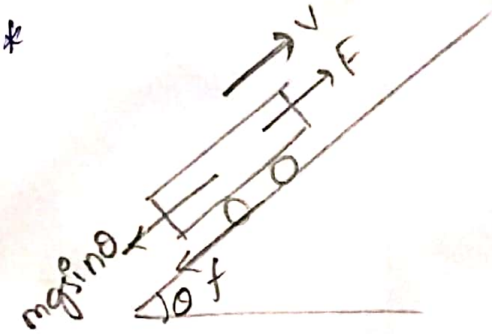
$\boxed{F = ma + R}$

Power of engine,

$\boxed{P = FV}$

$\boxed{P = (R + ma)v}$

*

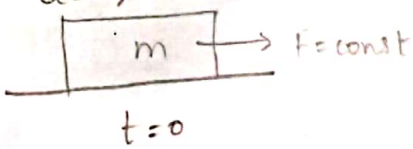


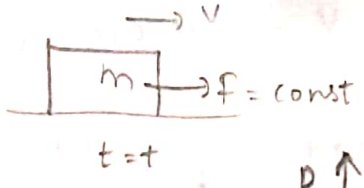
$P = FV$

$\boxed{P = (mg\sin\theta + f)V}$

*

① $u = 0$, rest





* $P_{inst} = F \cdot V = F \cdot at \Rightarrow \boxed{P \propto t}$



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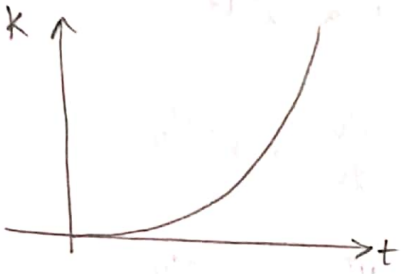
* $P_{avg} = \frac{1}{2} P_{inst}$

$\frac{W}{t} = \frac{1}{2} FV$

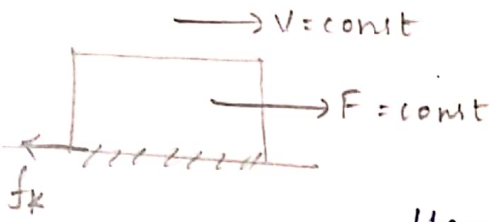
$W = \frac{1}{2} Fat^2$

$K_f - \cancel{K_i}^{0} = \frac{1}{2} Fat^2$

$\boxed{K.E = \frac{1}{2} Fat^2}$



(2)



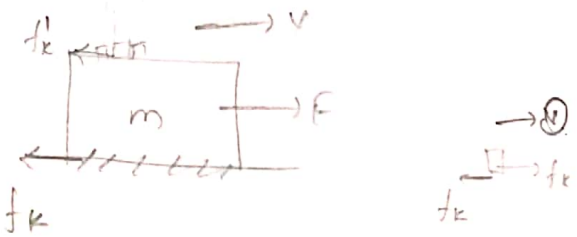
$P = FV$

$P = f_k V$

Here $P = const$

$\Rightarrow \boxed{P_{inst} = P_{avg}}$

*



$F_{extra} = V \frac{dm}{dt}$

$P = F \cdot V = V \cdot \frac{dm}{dt} \quad (V)$

$\boxed{P = V^2 \cdot \frac{dm}{dt}}$

→ A particle is moving along X-axis under the action of a force which varies it's position x as $F \propto \frac{1}{\sqrt{x}}$. Calculate the variation of power with x.

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$P_{inst} = F \cdot V$

$P = \frac{1}{2\sqrt{x}} \cdot \frac{dx}{}$

$a \propto x^{-1/4}$

$v \cdot \frac{dv}{dx} \propto x^{-1/4}$

$\frac{v^2}{2} \propto \frac{x^{3/4}}{3/4}$

$\frac{v^2}{2} = K\left(\frac{4}{3}\right) x^{3/4}$

$v^2 = K x^{3/4}$

$v = K x^{3/8}$

$P \propto x^{-1/4} \cdot x^{3/8}$

$P \propto x^{3-2/8}$

$\boxed{P \propto x^{1/8}}$

→ Power of the engine of a motor boat is 50hp. If resistance force F. of water increases with speed v of the boat according to $F = \alpha v$ Calculate Max speed of the motor boat is km/h

$P = F \cdot v$

$50 \times 746 = P$

For $v_{max} \Rightarrow a = 0$

$\Rightarrow F_{net} = 0$

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$F = F_{res} = 20V \times 10^3$.

$P = 20V \times 10^3 \times V$

$50 \times \frac{373}{746} = \frac{10}{20} \times 10^3 V^3_{max}$

$\frac{50 \times 373}{10^4} = V^3_{max}$

$\frac{50 \times 373}{100 \times 100} = V^3_{max}$

$V^3_{max} = \frac{373}{200}$

$V_{max} = \sqrt{\frac{373}{200}} = 1.36 \, m/s$

$V_{max} = 1.36 \times \frac{18}{5} \, km/hr$

$\boxed{V_{max} = 4.9 \, km/hr}$

An elevator can carry a max load of 1500kg is moving up with const speed of 2m/s. The friction force opposing the motion is 3000N. Calculate the min power developed by the motor to the elevator in kw & Hp ($g = 10 \, m/s^2$)

$P = 18000 \times 2$

$= 36000$

$\boxed{P = 36kW}$

$F = 3000 + 1500 \times 10$

$= 3000 + 15000$

$= 18000$

$P = \frac{36 \times 10^3}{746} = \frac{18 \times 10^3}{373} Hp = 48.3 Hp$

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